Precambrian Research 232 (2013) 119–139

Contents lists available at SciVerse ScienceDirect

Precambrian Research

journal homepage: www.elsevier.com/locate/precamres

The Paleoproterozoic Kaminak dykes, Hearne craton, western Churchill Province, Nunavut, Canada: Preliminary constraints on their age and petrogenesis

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ARTICLE INFO

Article history:
Received 11 May 2011
Received in revised form 24 May 2012
Accepted 4 June 2012
Available online 21 June 2012

Keywords:
Hearne craton
Kaminak dykes
Low-TiO_2 continental tholeiites
U–Pb baddeleyite geochronology
Lithogeochemistry
Nd isotopes
Petrogenesis

ABSTRACT

The NNW- to NNE-trending Kaminak diabase dykes, exposed in the Hearne craton of the western Churchill Province, Canada, are components of a Neoarchean–Paleoproterozoic LIP that forms an important pin in plate tectonic reconstructions impacting models for the assembly of the Nuna supercontinent. This study provides the first detailed information on the age, geochemistry and isotopic composition of these dykes.

A precise U–Pb baddeleyite date of 2497.6 ± 1.1 Ma, obtained for one dyke sample, provides the best estimate for the timing of Kaminak mafic magmatism. The dykes are tholeiitic basalts to basaltic andesites with consistent compositions over an areal extent of ~10,000 km^2. Major and compatible trace element variations indicate that the dykes were not in equilibrium with anhydrous peridotitic mantle and evolved through fractionation of Pl + Cpx ± Ol from parental melts in the deep crust. All dykes record remarkably similar, incompatible element enriched, closely parallel multi-element profiles with variable HFSE troughs, comparable to Phanerozoic low-TiO_2 continental tholeiites. Time-corrected Nd isotopic compositions overlap, or are slightly below bulk Earth ($\varepsilon_{Nd_{2500\,Ma}} = +0.4$ to -0.8), and yield T_{DM} values (2.75–2.87 Ga) comparable to those of igneous crust in the Hearne craton. Implied short residence times, along with the geological, geochemical and isotopic data, are interpreted to imply that parental magmas of the dykes were low-TiO_2 tholeiites generated via contamination of primary, depleted N-MORB-like magmas by low-degree, calc-alkaline lamprophyre melts in the continental lithospheric mantle. The melts homogenized in chambers near the Moho and underwent extensive fractional crystallization (Pl + Cpx ± Ol), and possibly minor assimilation of lower crust, prior to injection into the upper crust. Paucity of evidence for contemporaneous lithospheric rifting, homogeneous lithogeochemical and Nd isotopic compositions, and petrochemical variations distinct from the Matachewan and Mistassini swarms of the Superior Province suggest the Kaminak dykes are the magmatic products of a Late Archean–Early Paleoproterozoic mantle thermal anomaly that extended, but did not break apart the Hearne craton. The results of this study do not support direct correlation of the Kaminak and Matachewan or Mistassini dyke swarms. Further evaluation of the proposed linkage of the Hearne and Superior cratons prior to Nuna supercontinent assembly requires additional geochronological, lithogeochemical and radiogenic isotopic constraints.

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1. Introduction

Large Igneous Provinces (Coffin and Eldholm, 1994) are intrusive and extrusive magmatic expressions of catastrophic mantle events in the geological record that produce large volumes of dominantly tholeiitic magma (often > 1 million cubic kilometers) in relatively short periods of geologic time (commonly <10 m.y.). In regions of continental crust, Large Igneous Provinces (LIPs) can be recognized by the voluminous outpouring of continental flood basalts, giant dyke swarms, and clusters of associated mafic/ultramafic intrusions. In the Phanerozoic, there are numerous examples of short-lived outbursts of continental flood basalt volcanism (e.g., Columbia River, Deccan, Paraná, Karoo, Siberian trap, Central Atlantic Magmatic Province) and they generally follow this LIP definition.

In the Precambrian, however, deciphering the duration, extent and tectonic setting of LIPs is more challenging because many are substantially eroded and typically only feeder dykes, sills or layered mafic intrusions are preserved. Moreover, some LIPs are metamorphosed and deformed and it is possible that they can be fragmented by one or more periods of continental rifting. To

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http://dx.doi.org/10.1016/j.precamres.2012.06.002

add to these difficulties, geodynamic processes and lithospheric structure and thicknesses in the Precambrian are poorly understood. Despite these challenges, there are examples of relatively well-preserved Precambrian LIPs and, like their Phanerozoic counterparts, some are relatively short-lived events. For example, the 1.27 Ga Mackenzie Igneous Event in NW Canada impacted a region of stable continental crust, where the eruption of continental flood basalts (Dupuy et al., 1992), emplacement of a giant radiating dyke swarm (Mackenzie dyke swarm; Baragar et al., 1996; Heaman and LeCheminant, 1993; LeCheminant and Heaman, 1989; Schwab et al., 2004), and the emplacement of layered mafic/ultramafic intrusions (e.g., the Muskox intrusion; LeCheminant and Heaman, 1989; Mackie et al., 2009) occurred over a relatively short period between 1.272 and 1.265 Ga. In contrast, there is a growing number of Precambrian LIPs that appear to have a protracted history of tholeiitic magmatism that can last for up to 60 million years. Halls et al. (2008) linked the radiating nature of several Paleoproterozoic dyke swarms in the southern Superior Province to a common focal region, and interpreted the lack of evidence for apparent polar wander during the interval 2.125–2.067 Ga to indicate that the Marathon LIP represents an example of a long-lived mantle plume event with periodic pulses of mafic magmatism. An understanding of whether a Precambrian LIP is a short- or long-lived event and its tectonic setting are of vital importance if dyke swarms are to be appropriately used as piercing points in paleocontinental reconstructions (e.g., Bleeker and Ernst, 2006).

Other features of Precambrian LIPs that are poorly understood include the role of mantle plumes in their genesis, the composition and depth of the mantle source region where this magmatism originates, whether the magmatism accompanied continental rifting, and to what extent does contamination of these magmas occur during passage through the subcontinental lithospheric mantle and continental crust. Part of the reason for this is the preservation issue discussed above, but equally important is the dearth of geochemical and isotopic information available for ancient LIPs compared to the extensive literature available for Phanerozoic examples.

Precambrian mafic dyke swarms and, to a lesser extent, correlative eruptive flood basalts are a common feature of most Archean cratons (e.g., Boily and Ludden, 1991; Buchan et al., 1998, 2007; Cadman et al., 1993, 1997; Condie, 1997; Condie et al., 1987; Ernst et al., 2010 and references therein; French and Heaman, 2010; Halls and Fahrig, 1990; Heaman, 1997; Klausen et al., 2010; Kullerud et al., 2006; LeCheminant and van Breemen, 1994; LeCheminant et al., 1996; Maurice et al., 2009; Mertanen et al., 1999; Nelson et al., 1990; Paces and Bell, 1989; Phinney and Halls, 2001; Sheraton et al., 1989; Weaver and Tarney, 1981). In the Hearne craton of the western Churchill Province, at least four Proterozoic intracratonic mafic, mantle-derived suites are recognized, spanning > 250 m.y. of crustal evolution. The Hearne craton (Fig. 1) is host to: (1) the ca. 2.45 Ga Kaminak dykes (Fig. 2; Heaman, 1994); (2) the inferred correlative, but undated Spi Group basalts (Fig. 2; Beavon, 1976; Patterson, 1991; Sandeman and Ryan, 2008); (3) the <2.45 > 2.111 Ga Happotiyik Member of the Ameto Formation of the Hurwitz Group (Fig. 2; Sandeman et al., 2003) as well as correlatives in the Tavani segment (Fig. 1; Ridler, 1974); and (4) the 2111 Ma Griffin gabbro sills (Aspler et al., 2002; Heaman and LeCheminant, 1993).

This contribution outlines the field, petrographic, mineral and whole-rock geochemical, Nd-isotopic and conventional isotope dilution thermal ionization mass spectrometry (ID-TIMS) U–Pb baddeleyite geochronology of the Kaminak dykes, a suite of early Paleoproterozoic diabase dykes best exposed in the Kaminak Lake segment (Fig. 1) of the Hearne craton (Aspler and Chiarenzelli, 1996; Christie et al., 1975; Davidson, 1970; Eade, 1986; Hanmer et al., 2004). The petrogenesis of the dykes and the correlative Spi Group basalts (Sandeman and Ryan, 2008) are discussed and are compared to the low-TiO_2 Gramado basalts of the Paraná LIP (Garland et al., 1996; Peate and Hawkesworth, 1996). Finally, using geochemical data, we examine the postulated correlation between the Kaminak dykes described in this study and the Matachewan and Mistassini dyke swarms of the Superior craton (Bleeker, 2004; Bleeker and Ernst, 2006; Ernst and Buchan, 2010).

2. Regional geological setting

The western Churchill Province of the Canadian Shield is currently subdivided into the Rae craton to the north and northwest, the Hearne craton to the south, and the intervening Chesterfield block (Fig. 1; Berman et al., 2007). The southern part of the Hearne is poorly known, however, the occurrence of Paleo- to Mesoarchean inherited zircons and corresponding Paleo-Mesoarchean Sm–Nd T_{DM} ages (time of removal from depleted mantle; DePaolo, 1981a) from Paleoproterozoic granitoids in that area suggest that the present-day southernmost parts of the Hearne craton may contain an older, distinct cratonic fragment (van Breemen et al., 2005). The Hearne craton (Central Hearne subdomain or supracrustal belt of Hanmer et al., 2004), comprises a number of distinct supracrustal segments (Henik, Kaminak and Tavani) and collectively represents a large, predominantly volcanic–plutonic package of Neoarchean (ca. 2711–2666 Ma) rocks cropping-out over >300 km along a NE-striking segment of the former Ennadai–Rankin Greenstone Belt (Fig. 1; Aspler and Chiarenzelli, 1996; Wright, 1967). The Archean rocks are polydeformed (at least 2 episodes of deformation) and are cross-cut by the Kaminak dykes, a swarm of northwest- to northeast-trending (typically ~000° to 020°), plagioclase porphyritic to megacrystic diabase dykes best exposed in the Kaminak Lake area (Fig. 2; Christie et al., 1975; Davidson, 1970; Hanmer et al., 2004; Henderson, 1981). These portions of the Hearne craton have been favorably compared to the Abitibi belt of the southern Superior craton (Davis et al., 2004; Ridler and Shilts, 1974).

At Kaminak Lake, the Archean basement is unconformably and structurally overlain by a number of spatially discrete, synformal–antiformal segments of open-to-tightly folded Proterozoic rock sequences. These unconformable sedimentary sequences of poorly constrained age include the Spi Group (Beavon, 1976; Patterson, 1991; Sandeman and Ryan, 2008); the Hurwitz Group (Aspler and Chiarenzelli, 1996, 1997 and references therein), and two others, the MacKenzie Lake metasedimentary rocks (Aspler et al., 2000; Bell, 1968, 1971) and the Montgomery Lake Group (Aspler and Chiarenzelli, 1996; Eade, 1974) that are not discussed further herein. The Spi and Hurwitz Groups contain intracratonic mafic volcanic and hypabyssal rocks. Both groups exhibit broad similarities, in particular their coarse-grained, conglomeratic basal sections, the clastic sediment-rich nature of the strata, and the presence of plagioclase porphyritic basalts. The Spi Group, however, is much more restricted in extent, and is characterized by a NNE-trending (020°) foliation that parallels the stratigraphy and the poorly exposed basinal margins, but is oblique (~30° counterclockwise) to the orientation of the Hurwitz basin (~050°). These features collectively imply that the Spi Basin is both stratigraphically and structurally beneath the Hurwitz Group (see Sandeman and Ryan, 2008).

The Spi Group (Beavon, 1976; Patterson, 1991; Sandeman and Ryan, 2008) is exposed immediately west of Carr Lake (Fig. 2). It is a thin and areally restricted, dominantly sedimentary sequence containing a lowermost package of plagioclase porphyritic basaltic rocks termed the Spi Lake Formation. The basalts comprise up to 75–100 m of greenschist-facies, foliated, plagioclase porphyritic, green-grey flows with intercalated sandstone (Patterson, 1991). The Spi Lake Formation is overlain by matrix- to clast-supported, tonalite cobble conglomerate and sandstones of the Old Boot Formation (Patterson, 1991). Detrital zircon geochronology yields

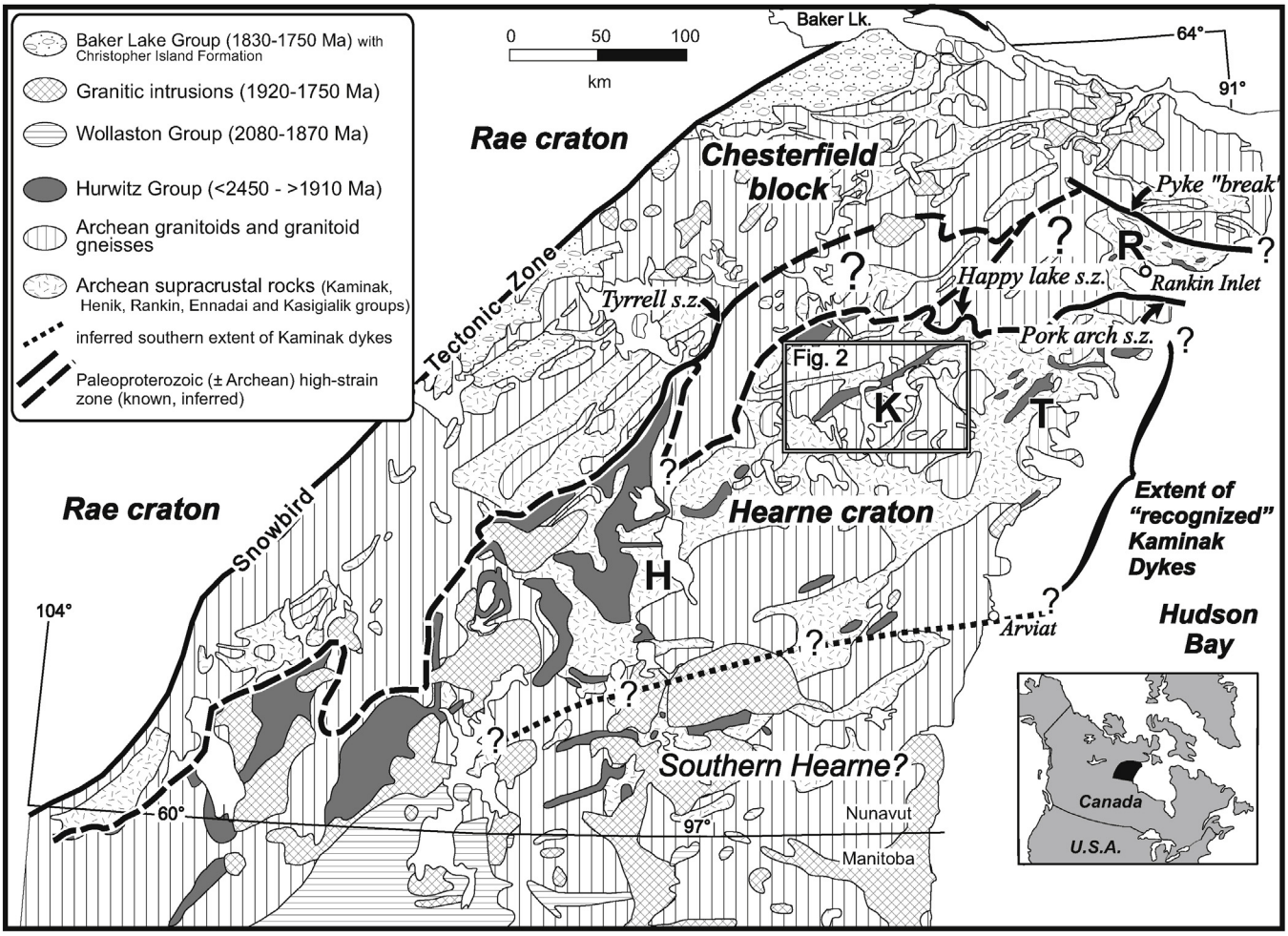


Fig. 1. Simplified geological map of the Hearne craton and environs. The region is divided into three major lithotectonic components, the Rae craton north and west of Snowbird Tectonic zone, the Hearne craton (central Hearne of Hanmer et al., 2004), and the intervening Chesterfield block (Berman et al., 2007; NW Hearne domain of Hanmer et al., 2004). Note the location of Fig. 2. Map modified after Aspler and Chiarenzelli (1996) and Stern and Berman (2000). Key: H – Henik belt; K – Kaminak belt; T – Tavani belt; R – Rankin Inlet belt. Inset at lower right shows location in North America.

a maximum depositional age of 2578 ± 20 Ma for the conglomerates of the Old Boot Formation (Rainbird et al., 2002). Based on their plagioclase-porphyritic character, apparent restriction to a NNE-trending basin developed in the Archean basement, and their spatial association with overlying coarse conglomerates, Beavon (1976) suggested that the Spi basalts were the eruptive equivalents of the Kaminak dykes, a proposal also advocated by Patterson (1991) and more recently supported by lithogeochemical and Nd isotopic data (Sandeman and Ryan, 2008).

Despite having been recognized since the 1950–1960s regional helicopter traverses (e.g., Wright, 1967), the full areal distribution and petrochemistry of the Kaminak dykes is poorly documented, in part because Quaternary cover increases dramatically both north and south of the Kaminak Lake area. Early studies (Christie et al., 1975; Davidson, 1970) documented the abundance, orientation and paleomagnetic character of the dykes in the Kaminak Lake area. Davidson (1970) noted a progressive northwards transition from fresh diabase dykes south of Kaminak Lake through a greenschist-facies corridor centered on the Hurwitz Group fold belt (Fig. 2). North of the Hurwitz Group, the dykes increase in metamorphic grade to amphibolite facies, preserve internal foliations, are transposed into the regional NE-trending structural fabric and, ultimately, are transformed into amphibolitic rafts in tonalitic orthogneiss (Aspler et al., 2000; Davidson, 1970; Hanmer et al., 2004). Although dykes have been observed or inferred up to 60 km to the north of Kaminak Lake, the extent and indeed occurrence of Kaminak dykes south of Kaminak Lake is poorly known (Davidson, 1970; Wright, 1967). Kaminak dykes are, however, documented in the Tavani area to the east (Park and Ralser, 1992; Peterson, 1997) and in the Henik Lake area to the southwest (Aspler and Chiarenzelli, 1997; Fig. 1).

The supracrustal and granitoid rocks exposed along the northern margin of intact Archean supracrustal rocks in the Henik–Kaminak–Tavani corridor exhibit evidence of penetrative, high-strain Paleoproterozoic deformation (thrusting?) and accompanying amphibolite grade metamorphism. Tectonometamorphic studies in the region (Berman et al., 2002) have demonstrated the development of ca. 1.84 Ga, middle amphibolite-facies grade, shallowly SE-dipping fabrics and metamorphic mineral assemblages in semipelitic rocks along the Happy lake shear zone (Happy lake s.z.; Fig. 1). These fabrics and mineral assemblages were interpreted to reflect the development of a regional, Paleoproterozoic, shallowly SE-dipping high-strain zone that acted as a major thrust fault, transporting the volcano-plutonic rocks of the Neoarchean Kaminak greenstone belt of the Hearne craton, northwards over granitoid and metasedimentary gneisses of unknown affiliation or age. Similarly, Tella et al. (1986) identified an east-west-trending high-strain zone (Pork arch s.z.) near the unexposed southern contact between the Rankin Inlet greenstone belt (R; Fig. 1) and strongly foliated granodiorite exposed to the south. Their similar orientations and

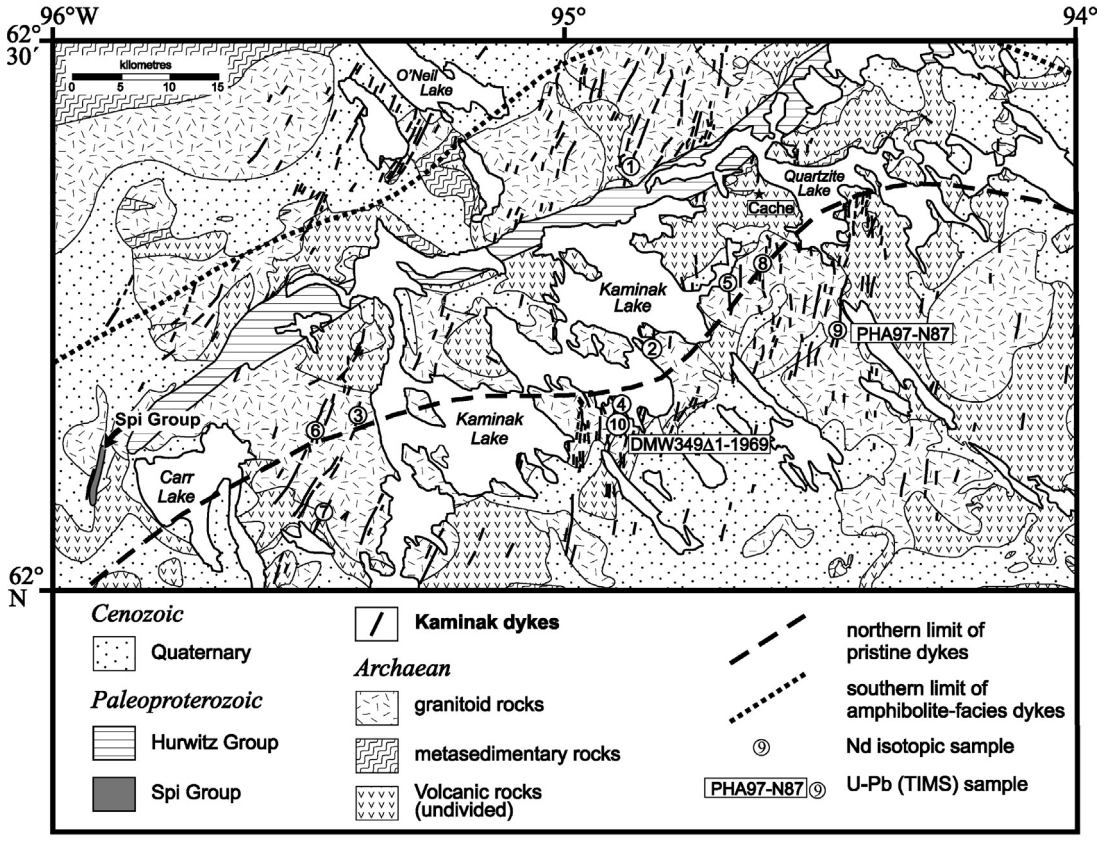


Fig. 2. Simplified geological map of the Carr–Kaminak–Quartzite Lake areas of the Hearne craton. Modified after Davidson (1970), Henderson (1981) and Hanmer et al. (2004). Note the geographical distribution and orientations of the Kaminak dykes and the locations of the U–Pb baddeleyite (#s 9, 10) and Sm–Nd (#s 1–8) isotopic samples.

the intensity of deformation along these high-strain zones suggest that they may be correlative. These structures are therefore of major regional significance and provide a rationale for the northwards increase in metamorphic grade and intensity of deformation in the Kaminak dykes. The presence of Kaminak dykes north of the Happy Lake shear zone is a subject of continued debate (see Berman et al., 2007). However, 15 km north of Rankin Inlet (Fig. 1), plagioclase porphyritic basalts associated with cobble conglomerate and deformed plagioclase megacrystic dykes have been interpreted as correlatives of the Spi Group basalts and Kaminak dykes, respectively (Carpenter, 2003).

3. Characteristics of the Kaminak dyke swarm

In the vicinity of Kaminak Lake, the Kaminak dykes comprise a swarm of typically steeply dipping, NW- to NE-trending (270–060° but typically 000–020°; median = 020; $n = 122$; Supplementary Appendix 1), black to brown-weathering diabase dykes that commonly form resistant ridges in the Archean host rocks (Fig. 2). They vary from rare, essentially aphyric or sparsely plagioclase porphyritic examples (Fig. 3A), to plagioclase porphyritic diabase, locally with spectacular plagioclase megacrystic examples (Fig. 3B). They are commonly more plagioclase porphyritic than described by Beavon (1976), having megacrysts ranging up to 10 cm in size. The dykes vary from <1 m to rarely more than 30 m in width (maximum = ~40 m) although dykes are typically 5–10 m wide. Individual dykes are segmented and laterally displaced by brittle faults (locally primary offsets?), consequently they can rarely be traced for more than a few kilometers. In the Carr–Kaminak–Quartzite Lake areas (Fig. 2), the dykes have well-preserved chilled margins and are generally undeformed, with the exception of a weak cleavage locally developed along their margins. They cut across the regional Archean foliation and metamorphic isograds, as well as significant Au-mineralization (Cache Deposit; Fig. 2) and thus provide important lithotectonic pins for discriminating the effects of Archean vs. Proterozoic tectonometamorphism.

The dykes are characterized by variable amounts of post-crystallization alteration, a result of the progressive increase in metamorphic grade north of the Kaminak Lake area (Fig. 2; Davidson, 1970; Henderson, 1981). They range from essentially pristine ophitic-textured diabase with minor saussuritization of plagioclase and uralitization of clinopyroxene (Fig. 3C), particularly in dykes exposed in the southern, low metamorphic grade domain, to dykes characterized by widespread, turbid saussuritization of plagioclase, uralitization of clinopyroxene and the growth of metamorphic chlorite, biotite, quartz and leucoxene (Fig. 3D). In the latter dykes, the primary textures are almost entirely destroyed and constituent minerals are extensively replaced. Subhedral plagioclase laths, varying from An_{66} to An_1 (Fig. 4a), are enclosed in oikocrysts of augite and ferroaugite, ranging from En_{51}–Fs_{39}–Wo_{10} to En_{43}–Fs_{17}–Wo_{40} (Fig. 4b). Typically the plagioclase ranges from An_{66} to An_{23}, and only phenocryst rims and matrix grains, particularly those in the metamorphosed dykes, yield albitic (An_{14}–An_1) compositions. Anhedral to locally subhedral titaniferous magnetite grains are paragenetically late and are interstitial to the clinopyroxene and plagioclase. Olivine is absent in all specimens and quartz was only observed in samples exhibiting extensive alteration and/or greenschist-facies metamorphism (Fig. 3d). Plagioclase phenocrysts vary from a few millimeters in fine-grained rocks up to 6 cm in length in megacrystic varieties, and exhibit cores of An_{64} with rims of An_{36}, comparable to matrix plagioclase. All subsequent mineral abbreviations are after Kretz (1983).

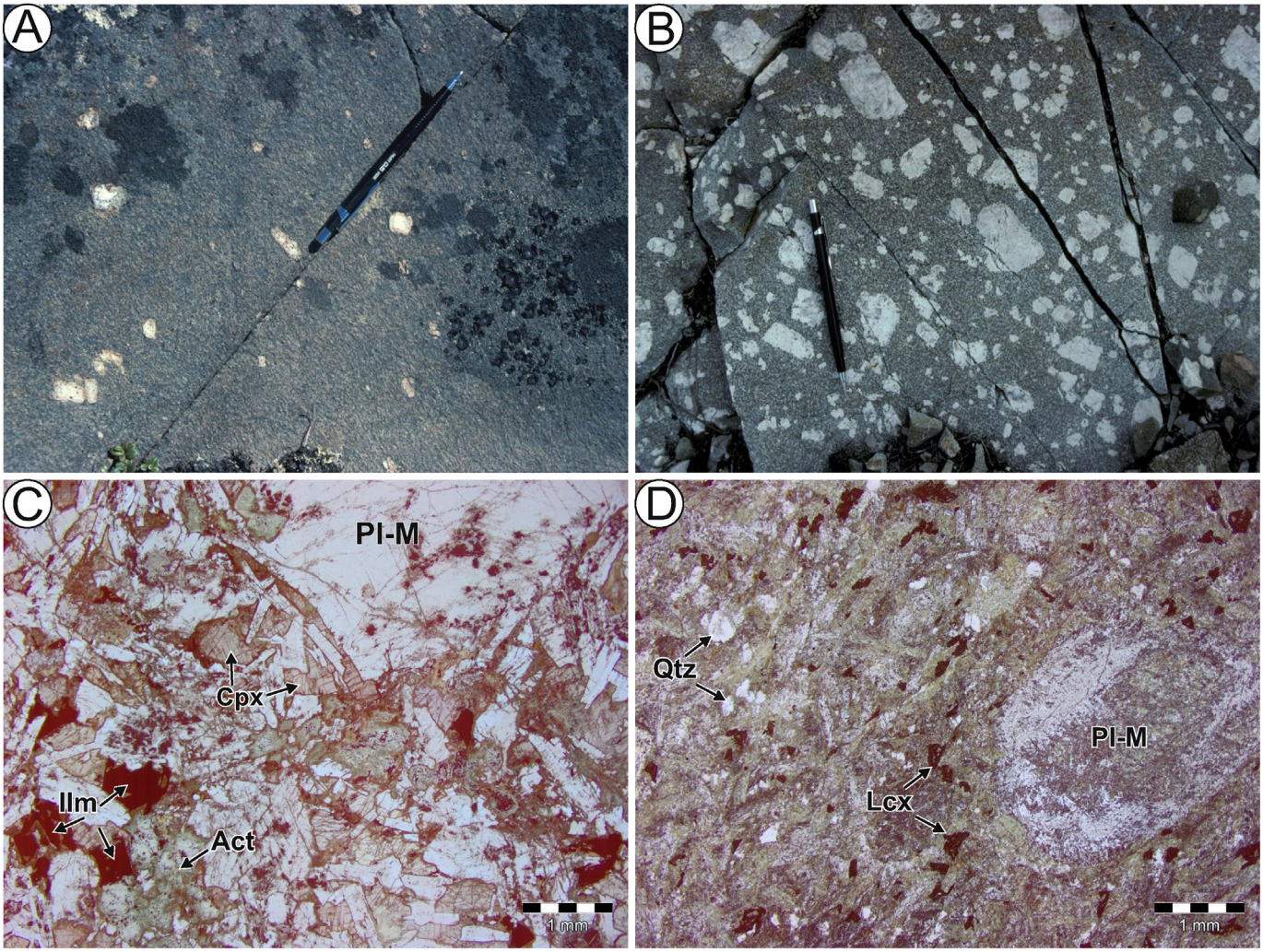


Fig. 3. Field photographs and photomicrographs of representative Kaminak dykes. (A) Sparsely plagioclase porphyritic dyke (PHA97-H108a); (B) plagioclase megacryst-rich dyke (PHA97-H520); (C) photomicrograph of relatively fresh Kaminak dyke (PHA97-N87), with a weakly altered plagioclase megacryst (Pl-M) surrounded by clinopyroxene (Cpx) locally overgrown by actinolite (Act), plagioclase laths and blocky anhedral ilmenite (Ilm); (D) photomicrograph of altered Kaminak dyke (PHA97-H108a), with a saussuritized plagioclase megacryst (Pl-M) surrounded by a matrix actinolite, albitic plagioclase, biotite, quartz (Qtz) and leucoxene (Lcx) after ilmenite.

4. U–Pb geochronology

Prior to this investigation, four samples of Kaminak diabase dykes had been processed for U–Pb geochronology, and a small number of tiny baddeleyite crystals (<40 μm in the longest dimension) were recovered from one sample (DMW349Δ1-1969), collected from a >45 m wide vertical dyke trending 014°. This sample was processed at the Jack Satterley Laboratory in Toronto through standard grinding (jaw crusher and disc mill) and mineral separation (magnetic and density) techniques and 10 baddeleyite crystals were recovered. Analytical methods for sample DMW349Δ1-1969 are given in Heaman and LeCheminant (1993), and preliminary results were reported by Heaman (1994). The U–Pb results for the single baddeleyite fraction, consisting of ten tiny, tan blades and fragments from sample DMW349Δ1-1969, are presented in Table 1 and on a concordia diagram in Fig. 5. Baddeleyite in this sample has high uranium (1266 ppm) and is quite discordant (7.1%), with a $^{207}Pb/^{206}Pb$ date of 2429.5 ± 2.0 Ma. This ca. 2430 Ma date is interpreted to represent a minimum estimate for the time of baddeleyite crystallization in this dyke.

A second attempt to find suitable Kaminak dyke samples for U–Pb geochronology was initiated in 2003 by first screening several geochemical samples collected during the western Churchill Province NATMAP Project (1997–2000) for U-bearing minerals such as baddeleyite or zircon in standard petrographic thin sections. Minuscule baddeleyite was identified in thin section in three samples (PHA97-N87, PHA97-H520 and PHA97-H492A). These three samples were processed at the Radiogenic Isotope Laboratory, University of Alberta, and approximately 200 g of powder from each sample were mixed in a water/calgon solution and then passed slowly over a Wilfley Table. The heavy mineral concentrate was then passed through magnetic (Frantz Isodynamic Separator) and density (Methylene Iodide) separation techniques. Approximately 20 tiny, light brown, baddeleyite blades and fragments were recovered from sample PHA97-N87, and these were separated into two, ten grain aliquots. Sample PHA97-N87 is a ~30 m wide, discontinuously exposed, weakly plagioclase porphyritic, medium-grained diabase dyke that trends 008/85°. The dyke has chill margins against, and cross-cuts medium-grained, hornblende-biotite tonalite of the Kaminak pluton (Davis et al., 2004; Sandeman et al., 2004a)

The two baddeleyite fractions from PHA97-N87 were cleaned in dilute (2 N) HNO_3, rinsed with Millipore water and acetone, and then deposited into TFE Teflon digestion vessels together with HF/8 N HNO_3 (5:1) and a measured amount of $^{205}Pb/^{235}U$ tracer solution. The fraction weights were estimated based on

Table 1
U–Pb Results for Kaminak Dykes, Hearne Province, Canada.

	Weight (μg)	Model Ages (Ma)											
		U (ppm)	Th (ppm)	Pb (ppm)	Th/U	TCPb (pg)	$^{206}Pb/^{204}Pb$	$^{206}Pb/^{238}U$	$^{207}Pb/^{235}U$	$^{207}Pb/^{206}Pb$	$^{206}Pb/^{238}U$	$^{207}Pb/^{235}U$	$^{207}Pb/^{206}Pb$	%Disc
PHA97-N87														
1b (6)	0.6	321	43	158	0.14	1.8	5288	0.4725 ± 7	10.684 ± 16	0.16399 ± 6	2494.5 ± 2.9	2496.0 ± 1.4	2497.2 ± 0.6	0.1
2b (6)	0.8	221	33	111	0.15	4.6	1151	0.4689 ± 10	10.618 ± 23	0.16420 ± 12	2479.0 ± 4.2	2490.2 ± 2.1	2499.4 ± 1.2	1.0
DMW349Δ1-1969														
3b (10)	0.5	1266	95	550	0.08	3.3	5210	0.4255 ± 6	9.243 ± 12	0.15755 ± 10	2285.3 ± 2.7	2362.3 ± 1.3	2429.5 ± 1.0	7.1

U–Pb baddeleyite data for Kaminak dykes PHA97-N87and DMW349Δ1-1969.
Notes. b – baddeleyite, number in parentheses corresponds to the number of grains analyzed. Atomic ratios corrected for blank (1 pg Pb; 0.5 pg U), mass discrimination (0.105%/amu Pb; 0.123%/amu U) and initial common Pb (Stacey and Kramers, 1975).
Th content estimated from the amount of ^{208}Pb present and $^{207}Pb/^{206}Pb$ date.
All errors in this table reported at 1σ.
TCPb – total common Pb.

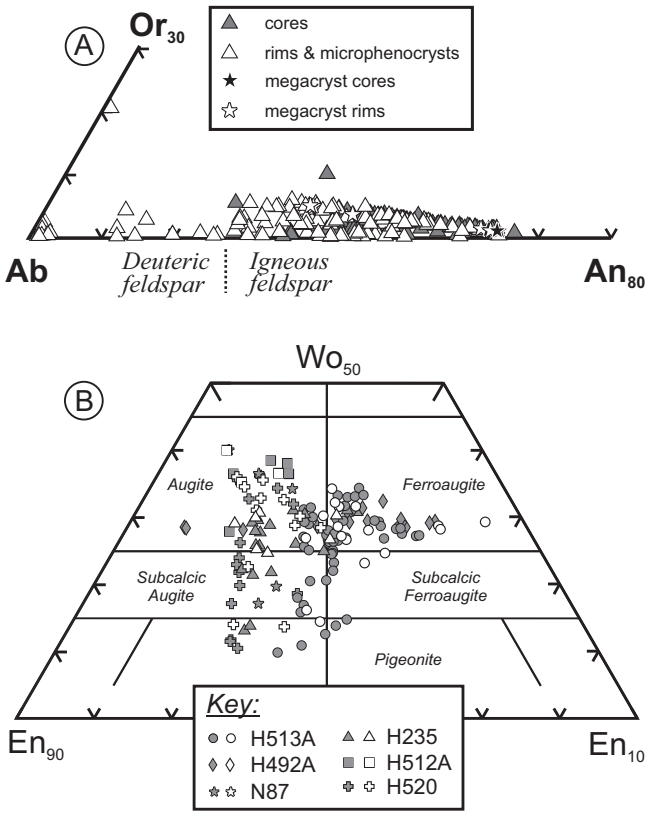


Fig. 4. (A) Plagioclase compositions of the Kaminak dykes plotted in Ab–Or–An space. Plagioclase grains between An_{14}–An_1 are interpreted as being deuteric or metamorphic in origin and (B) pyroxene compositions of the Kaminak dykes plotted in En–Wo–Fs space, showing the ranges in composition for clinopyroxene from six dykes. Filled symbols are grain cores, open symbols are rims.

the approximate volume of each crystal and the average density of baddeleyite (5.75 g/cc). The fractions were heated to 225 °C for 48 h, evaporated to dryness, and then converted to chloride form. Purification of U and Pb using conventional anion exchange chromatography was not conducted because of the small (<1 μg) sample weights, and it is our experience that the mass spectrometer run behavior of such small baddeleyite fractions is not

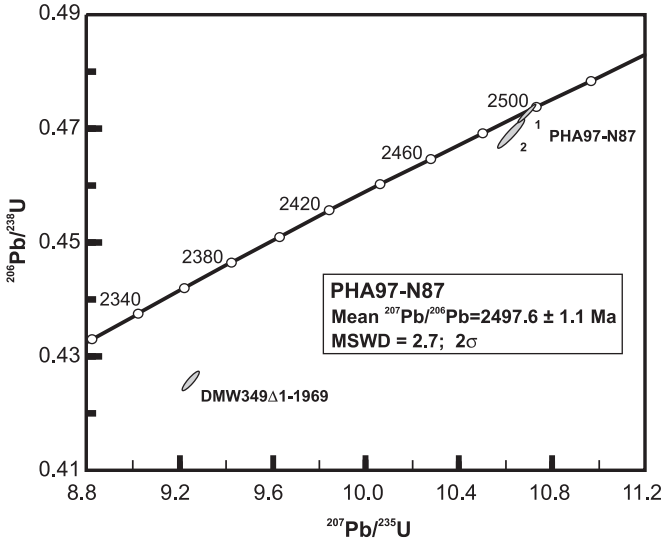


Fig. 5. U–Pb concordia plot showing two baddeleyite analyses obtained from Kaminak dyke PHA97-N87 and one baddeleyite analysis from Kaminak dyke DMW349Δ1-69.

Table 2
Lithogeochemistry for representative specimens of Kaminak dykes.

Sample	PHA97-N87	PHA97-H524B	PHA97-H522B	PHA97-H520	PHA97-H513A	PHA97-H512A	PHA97-H506B	PHA97-H505B	PHA97-H502B	PHA97-H493B
Rock type	Diabase	Diabase	Diabase	Diabase	Diabase	Diabase	Diabase	Diabase	Diabase	Diabase
Latitude	62.26068	62.28468	62.15245	62.11561	62.06304	62.13571	62.27897	62.15977	62.16470	62.05002
Longitude	−94.53749	−94.62333	−94.89703	−95.09370	−95.47446	−95.49388	−94.68006	−94.97117	−95.40490	−95.39505
Mg#	44.6	44.3	39.5	46.7	28.9	46.8	35.3	45.8	24.2	38.5
SiO_2	49.80	50.30	50.70	49.60	54.30	49.70	51.50	49.40	54.50	50.70
TiO_2	1.15	1.05	1.33	1.06	1.79	1.08	1.59	1.12	2.05	1.40
Al_2O_3	14.80	16.90	14.00	15.30	13.30	14.90	13.40	15.30	12.90	14.40
FeO	10.90	8.50	11.60	10.30	12.20	10.50	11.40	10.30	10.90	11.00
Fe_2O_3	3.00	3.00	2.30	3.10	2.60	2.50	3.10	3.00	3.70	3.10
MnO	0.20	0.16	0.20	0.18	0.19	0.19	0.19	0.17	0.17	0.18
MgO	6.14	4.99	5.00	6.44	3.31	6.30	4.34	6.17	2.55	4.85
CaO	9.00	9.43	8.22	8.65	7.38	8.79	8.13	8.39	6.75	8.13
Na_2O	2.40	2.50	2.20	2.50	2.70	2.60	2.50	2.20	2.40	2.50
K_2O	0.81	1.01	1.35	0.79	1.31	0.92	1.23	0.92	1.76	1.35
P_2O_5	0.13	0.13	0.17	0.12	0.27	0.13	0.19	0.13	0.31	0.18
LOI	0.70	1.60	1.10	0.90	0.60	1.10	1.40	1.90	0.90	1.30
Cr	83	90	64	60	26	91	33	59	24	59
Ni	120	120	84	190	54	120	84	170	31	110
Co	70	56	63	71	62	62	63	69	57	65
Sc	29	24	30	25	30	28	30	26	27	28
V	230	170	220	180	230	200	250	190	240	200
Cu	100	110	150	130	130	100	140	130	89	160
Pb	6	4	5	4	10	5	12	4	11	7
Zn	89	72	100	86	120	84	110	87	110	95
Mo	0.7	0.5	1.0	0.4	1.2	0.6	1.0	0.0	1.3	0.8
Sn	bd	bd	bd	bd	bd	bd	bd	bd	bd	bd
Tl	0.13	0.25	0.37	0.15	0.24	0.18	0.26	0.27	0.32	0.21
Cs	2.50	2.60	3.80	1.20	3.30	5.40	2.80	2.50	2.40	2.60
Ga	20	20	20	20	22	19	20	19	22	21
Ba	260	260	260	270	410	320	400	210	510	490
Rb	23	42	57	23	40	32	43	41	57	43
Sr	240	290	220	230	200	250	300	250	240	250
Hf	3.0	2.6	3.5	3.3	5.0	3.0	5.0	3.1	5.7	3.8
Ta	0.30	0.30	0.30	0.30	0.50	0.20	0.40	0.30	0.60	0.40
Nb	6.1	5.0	7.1	5.0	10.0	5.2	7.9	5.3	12.0	7.7
Y	22	20	26	20	33	22	28	21	35	26
Zr	94	85	118	90	186	94	147	93	216	132
Th	2.90	3.00	4.40	3.10	7.50	3.20	5.20	3.00	8.30	4.80
U	0.63	0.74	1.10	0.70	1.70	0.63	1.20	0.70	1.80	1.20
La	15.00	14.00	20.00	14.00	33.00	15.00	23.00	14.00	37.00	22.00
Ce	32.00	30.00	40.00	29.00	69.00	31.00	49.00	29.00	77.00	45.00
Pr	4.00	3.70	5.00	3.60	8.30	3.70	6.10	3.70	9.20	5.50
Nd	16.00	15.00	20.00	15.00	34.00	16.00	25.00	15.00	37.00	23.00
Sm	3.60	3.40	4.40	3.70	6.40	3.80	5.40	3.50	7.30	4.80
Eu	1.10	1.10	1.30	1.10	1.60	1.10	1.50	1.20	1.70	1.40
Gd	3.70	3.40	4.50	3.50	5.90	3.40	5.20	3.80	6.60	4.60
Tb	0.61	0.54	0.71	0.57	0.91	0.60	0.83	0.58	1.00	0.76
Dy	3.90	3.40	4.40	3.40	5.80	3.90	5.20	3.50	6.00	4.60
Ho	0.77	0.69	0.87	0.73	1.10	0.77	1.00	0.74	1.20	0.93
Er	2.10	1.90	2.50	1.90	3.10	2.20	2.80	2.00	3.20	2.50
Tm	0.32	0.27	0.36	0.31	0.48	0.33	0.43	0.30	0.49	0.39
Yb	2.00	2.00	2.40	2.10	3.00	2.10	2.80	2.00	3.20	2.50
Lu	0.33	0.30	0.36	0.31	0.48	0.37	0.43	0.28	0.49	0.41

Table 2 (*Continued*)

Sample	PHA97-N87	PHA97-H524B	PHA97-H522B	PHA97-H520	PHA97-H513A	PHA97-H512A	PHA97-H506B	PHA97-H505B	PHA97-H502B	PHA97-H493B
Rock type	Diabase	Diabase	Diabase	Diabase	Diabase	Diabase	Diabase	Diabase	Diabase	Diabase
Latitude	62.26068	62.28468	62.15245	62.11561	62.06304	62.13571	62.27897	62.15977	62.16470	62.05002
Longitude	−94.53749	−94.62333	−94.89703	−95.09370	−95.47446	−95.49388	−94.68006	−94.97117	−95.40490	−95.39505
$(La/Yb)_{CN}$	5.38	5.02	5.98	4.78	7.89	5.12	5.89	5.02	8.29	6.31
$(La/Sm)_{CN}$	2.69	2.66	2.93	2.44	3.33	2.55	2.75	2.58	3.27	2.96
$(Gd/Yb)_{CN}$	1.53	1.41	1.55	1.38	1.63	1.34	1.54	1.57	1.71	1.52
$(Th/Nb)_{CN}$	4.03	5.09	5.26	5.26	6.36	5.22	5.58	4.80	5.87	5.29
$(La/Nb)_{CN}$	2.55	2.91	2.92	2.91	3.43	2.99	3.02	2.74	3.20	2.97
Eu/Eu*	0.92	0.99	0.89	0.93	0.80	0.94	0.87	1.01	0.75	0.91
$\sum$REE	85.4	79.7	106.8	79.2	173.1	84.3	128.7	79.6	191.4	118.4

Sample	PHA97-H492A	PHA97-H235	PHA97-H108	PHA98-6-12b	PHA97-67-3	DMW435Δ1-1969	DMA400-1967	DMR155-1967	DMR113-1967	DMA298-1967
Rock type	Diabase	Diabase	Diabase	Diabase	Diabase	Diabase	Diabase	Diabase	Diabase	Diabase
Latitude	62.15552	62.21990	62.38333	61.56777	61.53266	62.57103	62.34434	62.41991	62.45979	62.85245
Longitude	−95.41391	−94.84457	−94.87859	−97.00704	−96.88374	−94.60772	−94.40004	−94.84891	−94.88773	−95.34520
Mg#	25.0	37.7	31.7	43.4	43.2	41.7	47.3	32.1	35.9	39.6
SiO_2	53.70	52.50	53.90	51.30	51.00	50.30	50.40	53.40	50.80	52.10
TiO_2	1.97	1.38	1.63	1.18	1.15	1.20	0.95	1.71	1.50	0.90
Al_2O_3	12.70	14.00	13.50	14.90	14.70	14.50	15.50	13.30	14.10	16.70
FeO	12.10	11.30	10.20	10.30	10.40	10.60	9.90	10.20	10.40	7.90
Fe_2O_3	2.60	2.50	3.80	3.00	3.10	3.60	3.10	4.10	4.00	3.50
MnO	0.18	0.19	0.17	0.19	0.20	0.20	0.19	0.19	0.19	0.17
MgO	2.70	4.60	3.55	5.57	5.64	5.55	6.40	3.67	4.41	4.04
CaO	6.36	8.49	6.55	9.12	9.54	9.28	9.41	7.30	8.49	10.73
Na_2O	2.80	2.20	2.80	2.70	2.80	2.60	2.60	3.00	2.60	2.70
K_2O	1.96	1.49	2.05	0.95	0.37	0.92	0.85	0.94	1.23	0.68
P_2O_5	0.29	0.18	0.23	0.14	0.13	0.14	0.11	0.23	0.20	0.12
LOI	0.50	0.60	1.20	0.60	0.40	0.70	0.90	1.40	1.60	0.80
Cr	27	46	39	96	82	84	67	33	50	63
Ni	31	84	56	88	100	82	150	57	110	52
Co	54	60	57	55	60	72	60	52	62	35
Sc	28	29	29	31	30	32	29	29	27	25
V	240	220	220	220	210	250	180	210	190	210
Cu	89	120	130	140	110	92	180	110	150	48
Pb	10	4	9	6	7	6	5	7	9	9
Zn	120	97	110	110	110	110	89	130	120	62
Mo	1.4	bd	0.9	0.5	0.3	0.4	2.3	1.0	0.6	1.0
Sn	bd	bd	bd	0.9	1.0	0.9	1.0	2.0	1.3	1.2
Tl	0.39	0.30	0.48	0.17	0.08	0.17	0.16	0.21	0.25	0.06
Cs	3.7	5.5	6.8	1.8	1.0	1.9	4.4	1.6	3.8	1.5
Ga	22	21	21	20	20	21	19	21	21	25
Ba	470	410	430	352	80	235	262	268	377	101
Rb	58	47	77	26	9	26	26	31	40	8.7
Sr	220	200	300	220	250	200	220	230	260	610
Hf	5.8	3.8	4.2	2.6	2.6	2.3	1.9	4.2	3.3	1.9
Ta	0.50	0.50	0.50	0.35	0.37	0.40	0.29	0.62	0.52	0.30
Nb	12.0	7.3	9.1	5.9	5.8	6.3	4.5	9.5	7.8	4.9
Y	34	28	31	24	24	24	18	31	28	24
Zr	206	144	178	106	107	93	82	170	140	84
Th	8.6	4.7	6.6	3.5	3.4	3.3	2.6	6.7	5.1	2.7
U	1.8	1.0	1.3	0.69	0.56	0.73	0.72	1.5	1.2	0.77
La	35.0	23.0	29.0	12.0	17.0	17.0	12.0	29.0	24.0	21.0
Ce	72.0	47.0	60.0	25.0	35.0	34.0	25.0	57.0	49.0	38.0
Pr	8.8	5.7	7.2	3.3	4.4	4.4	3.2	7.1	6.1	4.8

Nd	36.0	23.0	30.0	14.0	18.0	18.0	13.0	28.0	24.0		19.0
Sm	7.6	5.0	6.3	3.4	3.9	3.9	2.8	6.0	5.1		3.9
Eu	1.7	1.5	1.5	1.0	1.2	1.2	0.89	1.6	1.4		1.6
Gd	6.8	5.0	5.8	3.7	3.9	4.0	2.8	5.7	5.0		3.8
Tb	1.1	0.81	0.87	0.66	0.68	0.70	0.49	0.90	0.83		0.65
Dy	6.2	4.4	5.3	3.6	3.6	3.7	2.7	4.9	4.4		3.7
Ho	1.2	0.97	1.1	0.80	0.81	0.80	0.59	1.0	0.94		0.82
Er	3.3	2.6	2.9	2.3	2.3	2.3	1.6	2.9	2.6		2.2
Tm	0.48	0.39	0.44	0.34	0.33	0.33	0.25	0.44	0.39		0.32
Yb	3.2	2.6	2.8	2.3	2.3	2.2	1.7	3.0	2.6		2.1
Lu	0.51	0.39	0.46	0.34	0.33	0.35	0.25	0.44	0.38		0.31
$(La/Yb)_{CN}$	7.85	6.35	7.43	3.74	5.30	5.54	5.06	6.93	6.62		7.17
$(La/Sm)_{CN}$	2.97	2.97	2.97	2.28	2.81	2.81	2.77	3.12	3.04		3.48
$(Gd/Yb)_{CN}$	1.76	1.59	1.71	1.33	1.40	1.50	1.36	1.57	1.59		1.50
$(Th/Nb)_{CN}$	6.08	5.46	6.15	5.03	4.97	4.44	4.90	5.98	5.55		4.67
$(La/Nb)_{CN}$	3.03	3.27	3.31	2.11	3.04	2.80	2.77	3.17	3.19		4.45
Eu/Eu*	0.72	0.92	0.76	0.86	0.94	0.93	0.97	0.84	0.85		1.27
$\sum$REE	183.9	122.4	153.7	72.7	93.8	92.9	67.3	148.0	126.7		102.2

Whole-rock lithogeochemical data for 20 representative Kaminak dykes.
Major elements in weight % oxide, trace elements in ppm. Trace elements Co, Cr, Cu, Ni, Sc, V and Zn were obtained by ICP-ES, Ba by XRF. All remaining trace elements by ICP-MS. All latitude and longitude coordinates given in NAD27 datum. Key: LOI = loss on ignition; Mg# = molecular $(MgO/MgO + FeO_T) \times 100$; bd = below detection.

jeopardized by omitting the U and Pb purification steps. The U and Pb isotopic compositions were determined using a VG354 thermal ionization mass spectrometer operating in single Daly photomultiplier detector mode at the University of Alberta. U and Pb were loaded together onto single outgassed Re filaments in a mixture of 0.1 N H_3PO_4/Si gel, and data were collected at temperatures between 1400–1550 °C (Pb) and 1500–1600 °C (U). The age calculation was performed using Isoplot (Ludwig, 2003), utilizing the uranium decay constants (^{238}U-1.55125 × 10^{-10} yr.$^{-1}$ and ^{235}U-9.8485 × 10^{-10} yr.$^{-1}$) determined by Jaffey et al. (1971). The isotopic composition of common Pb in these analyses in excess of the estimated procedural blank of 1 pg was determined using the two-stage average crustal Pb evolution model of Stacey and Kramers (1975).

The U–Pb results for the two baddeleyite fractions from sample PHA97-N87 are presented in Table 1 (with one sigma errors) and on a concordia diagram in Fig. 5. Both fractions have slightly elevated Th/U of ~0.15 (most baddeleyite analyses have Th/U < 0.1) and low total common Pb contents (<5 pg). Baddeleyite fraction #1 overlaps the concordia curve (0.1% discordant; Fig. 5), with a ^{207}Pb/^{206}Pb date of 2497.2 ± 1.2 Ma (2σ). Baddeleyite fraction #2 is slightly more discordant (1.0%), but has a similar ^{207}Pb/^{206}Pb date of 2499.4 ± 2.4 Ma (2σ) that, within analytical uncertainty, agrees with the result obtained from fraction #1. The weighted mean ^{207}Pb/^{206}Pb date of 2497.6 ± 1.1 Ma (MSWD = 2.7) is interpreted as the best estimate for the time of baddeleyite crystallization in this sample and a relatively accurate estimate for the time of dyke PHA98-N87 emplacement.

The U–Pb results from the two analyzed dyke samples could indicate contemporaneous emplacement or that the Kaminak dyke swarm had a protracted emplacement history spanning ≤70 million years. For sample DMW349Δ1-1969 (7.1% discordant and a ^{207}Pb/^{206}Pb date of 2429.5 ± 2.0 Ma), a disturbance in the U–Pb system at 1100 Ma would result in an upper intercept date of 2496 Ma, similar to the emplacement age obtained for dyke sample PHA97-N87. The weighted mean ^{207}Pb/^{206}Pb date of 2497.6 ± 1.1 Ma (MSWD = 2.7) for dyke PHA98-N87 is far more robust than that for DMW349Δ1-1969 and is the only well constrained emplacement age for the Kaminak dykes.

5. Geochemistry

Major element analyses for 67 Kaminak dykes are reported, representing the first robust lithogeochemical dataset for these rocks. These include 20 specimens collected by the authors (prefixes PHA and LAA in Table 2) and 47 samples obtained from the Geological Survey of Canada archival collection of W. Fahrig and A. Davidson (prefixes FA or DM, etc.). A subset of 38 dykes was analyzed for trace and rare earth element (REE) contents and a smaller subset of 9 dykes was analyzed for Nd isotopic compositions. Representative dyke analyses are shown in Table 2 and the complete geochemical dataset is available in Supplementary Appendix B (online data repository). Duplicate analyses and precious metal data for some of these rock samples are presented in Ernst and Buchan (2010). Dyke samples having the PHA prefix were obtained from fine-to-medium-grained diabase, 50 cm to 1 m inside the chilled margins of the dykes and avoided, where possible, zones that may have concentrated plagioclase. This protocol is considered to minimize both the effect of plagioclase phenocryst-megacryst accumulation as well as any possible alteration or wall rock contamination within the outer chilled margins of the dykes.

The Kaminak dykes are compared to: (1) the Spi basalts, the inferred extrusive equivalents to the Kaminak dykes (Sandeman and Ryan, 2008); (2) the Matachewan dykes of the Superior craton (Boily and Ludden, 1991; Condie et al., 1987; Ernst and Buchan, 2010; Nelson et al., 1990; Phinney and Halls, 2001); (3) the

Table 3
Nd isotopic data for 9 samples of Kaminak dykes.

Sample	$^{147}Sm/^{144}Nd$ (measured)	$^{143}Nd/^{144}Nd$ (measured)	Location (Fig. 2)	$^{143}Nd/^{144}Nd$ (at t = 2500 Ma)	ε_{Nd} (at t = 2500 Ma)	ε_{Nd} (at t = 0)	T_{DM} (Ma)
PHA97-H108	0.1257	0.511472 (5)	1	0.509408	0.00	−22.75	2748
PHA97-H235	0.1297	0.511492 (6)	2	0.509363	−0.89	−22.35	2845
PHA97-H502B	0.1235	0.511391 (5)	3	0.509363	−0.89	−24.33	2822
PHA97-H505B	0.1357	0.511592 (3)	4	0.509364	−0.87	−20.40	2873
PHA97-H506B	0.1294	0.511499 (11)	5	0.509375	−0.66	−22.22	2823
PHA97-H512A	0.1353	0.511621 (9)	6	0.509400	−0.17	−19.84	2796
PHA97-H513A	0.1263	0.511448 (5)	7	0.509374	−0.67	−23.21	2812
PHA97-H524B	0.1331	0.511568 (7)	8	0.509383	−0.50	−20.87	2822
PHA97-N87	0.1363	0.511663 (12)	9	0.509426	0.34	−19.02	2746

Mistassini dykes of the Superior craton (Buchan et al., 1998; Ernst and Buchan, 2010; Heaman, 1994; (4) the low-TiO_2, tholeiitic, Gramado-type continental flood basalts of the Phanerozoic Paraná LIP (Garland et al., 1996; Peate and Hawkesworth, 1996); and (5) a field (2000+ analyses) for global mid-ocean ridge basalts (major elements), or an average value for the trace elements (N-MORB; Lehnert et al., 2000; http://www.petdb.org/; Sun and McDonough, 1989), respectively. N-MORB is shown for comparison as it is considered to represent low-P, asthenospheric mantle melts and derivatives generated without the influence of continental lithosphere. The Matachewan and the Mistassini dykes are shown for comparison because these have been interpreted as correlatives of the Kaminak dykes and therefore have significant implications for pan-Laurentia correlations (cf., Bleeker, 2004; Bleeker and Ernst, 2006). The Gramado-type Paraná basalts represent well-documented Mesozoic examples of a low-TiO_2 continental flood basalt sequence inferred to have been derived from the continental lithospheric mantle and have petrochemical characteristics very similar to the Kaminak dykes.

5.1. Analytical methods

All major, trace and REE data were obtained at the Geological Survey of Canada (GSC), Ottawa, Ontario. For the 20 samples collected by the authors, approximately 1.5 kg of each rock sample was crushed to chips in a Braun jaw crusher, and a 50 g split of each was pulverized to a fine powder in an agate ring mill. The archival samples of A. Davidson and W. Fahrig were reanalyzed using pre-existing whole-rock powders for which the processing methodology is not known. For all samples, the major elements were analyzed by X-ray fluorescence (XRF) analysis of fused discs. FeO was analyzed by titration. Volatile contents, expressed as loss on ignition (LOI) were determined gravimetrically. Ba was determined via XRF analysis of fused discs. The trace elements Co, Cr, Cu, Ni, Sc, V and Zn were analyzed by inductively coupled plasma-emission spectrometry (ICP-ES), and all other trace elements were determined through inductively coupled plasma-mass spectrometry (ICP-MS). Analytical uncertainties of the data, based on the analysis of duplicates and of reference materials (SY-2, BHVO-1, PCC-1 and replicate analysis of 5 internal standards) were calculated at <5% relative percent for the major elements, <10% for those trace elements determined by XRF, <10% for ICP-ES analyses and <5% for ICP-MS analyses. Some discrepancy in the accuracy of the ICPMS data for the elements Zr and Ta, however, was noted between analytical batches at the GSC Laboratory. These data yield spurious Nb/Ta and Zr/Hf ratios, leading to artificially complicated Zr–Hf and Nb–Ta behavior.

The Sm–Nd isotopic analyses were conducted at the GSC Radiogenic Isotope Laboratory, with analytical procedures of Thériault (1990) and Sandeman et al. (2004a,b). ε_{Nd} values were calculated with a present-day CHUR (chondritic uniform reservoir) composition of $^{143}Nd/^{144}Nd = 0.512638$ and $^{147}Sm/^{144}Nd = 0.1967$, using the decay constant of Lugmair and Marti (1978). T_{DM} values were calculated using the model of DePaolo (1981a). The 2σ reproducibility of the ε_{Nd} values is $\pm 0.5\varepsilon$ units. The Sm–Nd isotopic data are given in Table 3.

5.2. Classification

Low temperature alteration and metamorphism of mafic rocks, in particular ancient examples, may result in significant scatter of lithogeochemical data, particularly for the more mobile major and large ion lithophile (LILE) trace elements (cf. Middelburg et al., 1988; Pearce and Cann, 1973). However, as the Kaminak dykes exhibit strong linear interdependence of most elements, with the exception of some LILE (K_2O, Ba, Cs, Rb) it is inferred that the abundances of most elements have not been significantly modified since their emplacement in the Paleoproterozoic, and thus represent primary igneous concentrations.

In terms of both major and immobile incompatible trace element contents, the Kaminak dykes comprise basalts and basaltic andesites (Fig. 6; LeBas et al., 1986; Pearce, 1996). They are subalkaline, and exhibit FeO_T and TiO_2 tholeiitic enrichment trends (Fig. 7A) in both mobile and immobile major elements (Irvine and Barager, 1971; Jensen, 1976; Miyashiro, 1974). The dykes are both quartz and olivine normative tholeiites (Fig. 7B), and plot between the 1 atm (1 atm = 101.325 kPa) Ol + Cpx + Pl + liquid cotectic (line 1, Fig. 7B) and the deep crustal, Pl + Cpx ± Ol ± Mag cotectic (line 2, Fig. 7B), representing a basaltic andesite undergoing fractionation at the base of the crust (Thompson et al., 1983). These observations suggest the Kaminak dykes represent fractionated, modestly silica-rich basaltic magmas that ultimately equilibrated under crustal conditions between the two cotectic curves.

5.3. Major and compatible trace element variations

Selected major and trace elements are plotted against Mg# (molecular $(MgO/MgO + FeO_T \times 100)$ as a differentiation index (Fig. 8). The major elements Al_2O_3 and CaO and the trace elements Co, Cr and Ni decrease with fractionation (decreasing Mg#, MgO and increasing Zr), observations that collectively indicate Pl + Cpx ± Ol were fractionated from the parental magmas. Sr exhibits significant scatter, but appears to increase with fractionation, as does Eu, variations that support plagioclase crystallization. Petrographic observations, along with CaO, P_2O_5, TiO_2 and FeO_T interrelationships, indicate that apatite, ilmenite and magnetite crystallization was minor, and consistent with thin section observations that these phases are paragenetically late and thus did not significantly fractionate the derivative melts with respect to these elements. Most trace elements, and particularly the high field strength (HFSE) and REE, exhibit negative co-linear variation with Mg#, yielding highly correlated elemental variations (correlations with $r > 0.93$). For example, plots of $\log TiO_2$ and $\log P_2O_5$ vs. $\log Zr$ (Fig. 9; after Pearce and Norry, 1979) demonstrate that the

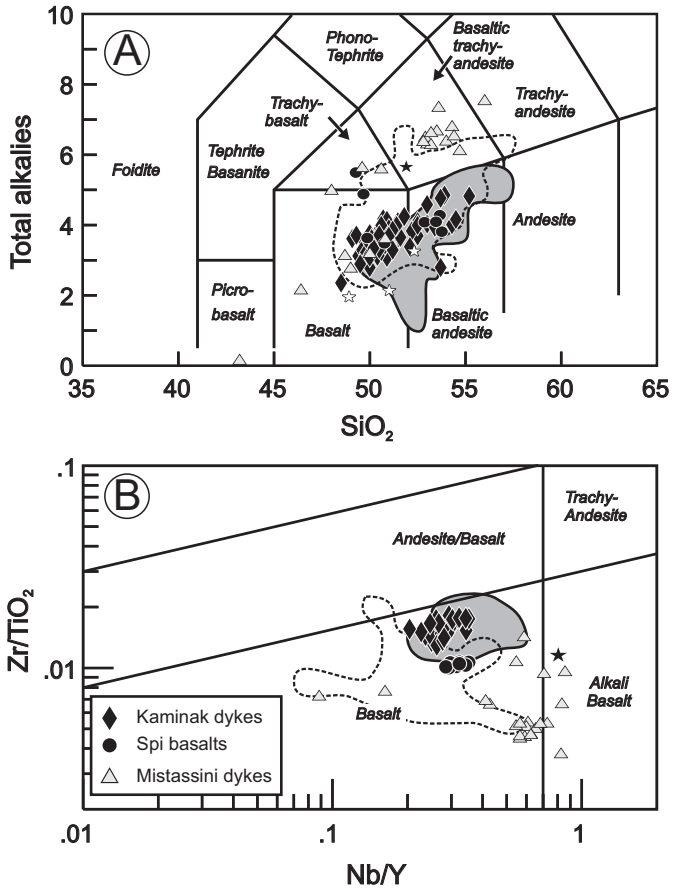


Fig. 6. (A) Total alkalies vs. SiO_2 (LeBas et al., 1986) and (B) Zr/TiO_2 vs. Nb/Y diagrams (Pearce, 1996) for the Kaminak dykes and Spi Group basalts. Shown for comparison are Mistassini dykes (Ernst and Buchan, 2010), the type-2 Matachewan dykes (dashed field; Nelson et al., 1990; Condie et al., 1987; Phinney and Halls, 2001; Ernst and Buchan, 2010), the type-1 Matachewan dykes (open stars), a type-3 Matachewan dyke (filled star), and the low-TiO_2 Gramado basalts of the Paraná LIP (shaded field; Turner and Hawkesworth, 1995; Garland et al., 1996; Peate and Hawkesworth, 1996).

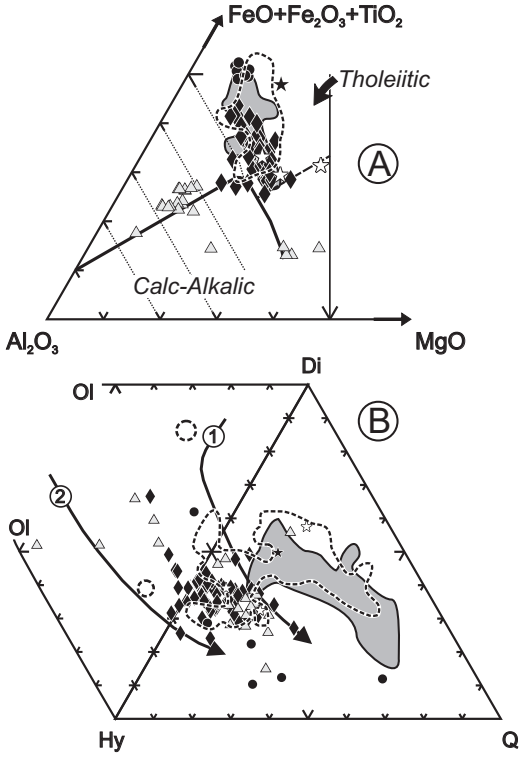


Fig. 7. (A) Jensen cation plot (Jensen, 1976) and (B) normative diopside, olivine, hypersthene and quartz (CIPW, wt.%) plot of the Kaminak dykes, Spi basalts. Fields, symbols and references as in Fig. 6. In (B), line 1 represents the cotectic curve for basaltic liquids in equilibrium with Ol + Pl + Cpx at 1 atmosphere; line 2 is the cotectic curve for basaltic liquids in equilibrium with Pl + Cpx ± Ol ± Mag deep in the continental crust (after Thompson et al., 1983).

Kaminak dykes exhibit steep, well-correlated (correlation coefficient $r = 0.966$ and 0.98, respectively), positive co-linear variations of these elements, also indicating the control on fractionation exerted by crystallization of the assemblage Pl + Cpx ± Ol without significant crystallization of magnetite, apatite or any other HFSE- or REE-bearing accessory phase.

5.4. Multi-element diagrams

N-MORB-normalized (Sun and McDonough, 1989) multi-element plots for the Kaminak dykes are shown in Fig. 10A and compared to petrochemical types—1, -2, -3 of the Matachewan swarm (Boily and Ludden, 1991; Condie et al., 1987; Ernst and Buchan, 2010; Nelson et al., 1990; Phinney and Halls, 2001). Fig. 10B shows data for the Spi Group basalts, the low-TiO_2 Gramado basalts of the Paraná LIP and the Mistassini dykes for comparison. The Kaminak dykes exhibit a remarkable coherency in their REE and incompatible trace element geochemistry, defining a series of parallel, closely spaced patterns. Their multi-element patterns exhibit negative slopes, strong light rare earth element (LREE) enrichment [$(La/Yb)_N = 3.74–8.29$; mean = 5.86, $n = 35$], and have pronounced troughs at Nb, Ta, P and Ti and less pronounced troughs at Zr–Hf. Negative Eu anomalies are also less pronounced ($Eu/Eu^* = 0.72–1.05$; Taylor and McLennan, 1985). The magnitude of the Eu trough (Eu/Eu^*; Fig. 11) increases with fractionation (lower Mg#'s and higher Zr abundances). Sr is also characterized by troughs, although one sample (DMA298-1967) has significant positive Eu and Sr anomalies. Ba, Zr and Hf exhibit minor troughs that are variable in magnitude.

5.5. Sm–Nd isotopic data

Sm–Nd isotopic data for 9 Kaminak dykes are presented in Table 3 and include time corrected ($t = 2.5$ Ga) $^{144}Nd/^{143}Nd$ ratios, ε_{Nd_t} values and depleted mantle model ages (T_{DM}'s). All samples yield present-day $^{144}Nd/^{143}Nd$ ratios ranging from 0.511391 to 0.511663, corresponding to ε_{Nd_t} of +0.4 to −0.8. These cluster around and extend to values less than the bulk Earth at 2.5 Ga. The dykes all have $^{147}Sm/^{144}Nd < 0.14$ and yield robust depleted mantle model dates that vary between 2.75 and 2.87 Ga (mean = 2.81 Ga).

6. Discussion

Interpretation of the origin of ancient continental flood basalts, such as the Kaminak dykes, are based on salient geological and geochemical observations derived from Phanerozoic continental flood basalt provinces (Anderson et al., 1992; Arndt and Christensen, 1992; Bondre et al., 2006; Cox, 1980; Gallagher and Hawkesworth, 1992; Garfunkel, 2008; Garland et al., 1996; Gibson et al., 1996; Hawkesworth et al., 1990; Hergt et al., 1991; Melluso et al., 2006; Peate and Hawkesworth, 1996; Ray et al., 2007). The petrogenesis of Phanerozoic continental flood basalts (CFB) has received considerable attention, but despite a large amount of research, their origin remains controversial (e.g., Foulger et al., 2005; Mahoney and Coffin, 1997). The controversy surrounds the role of

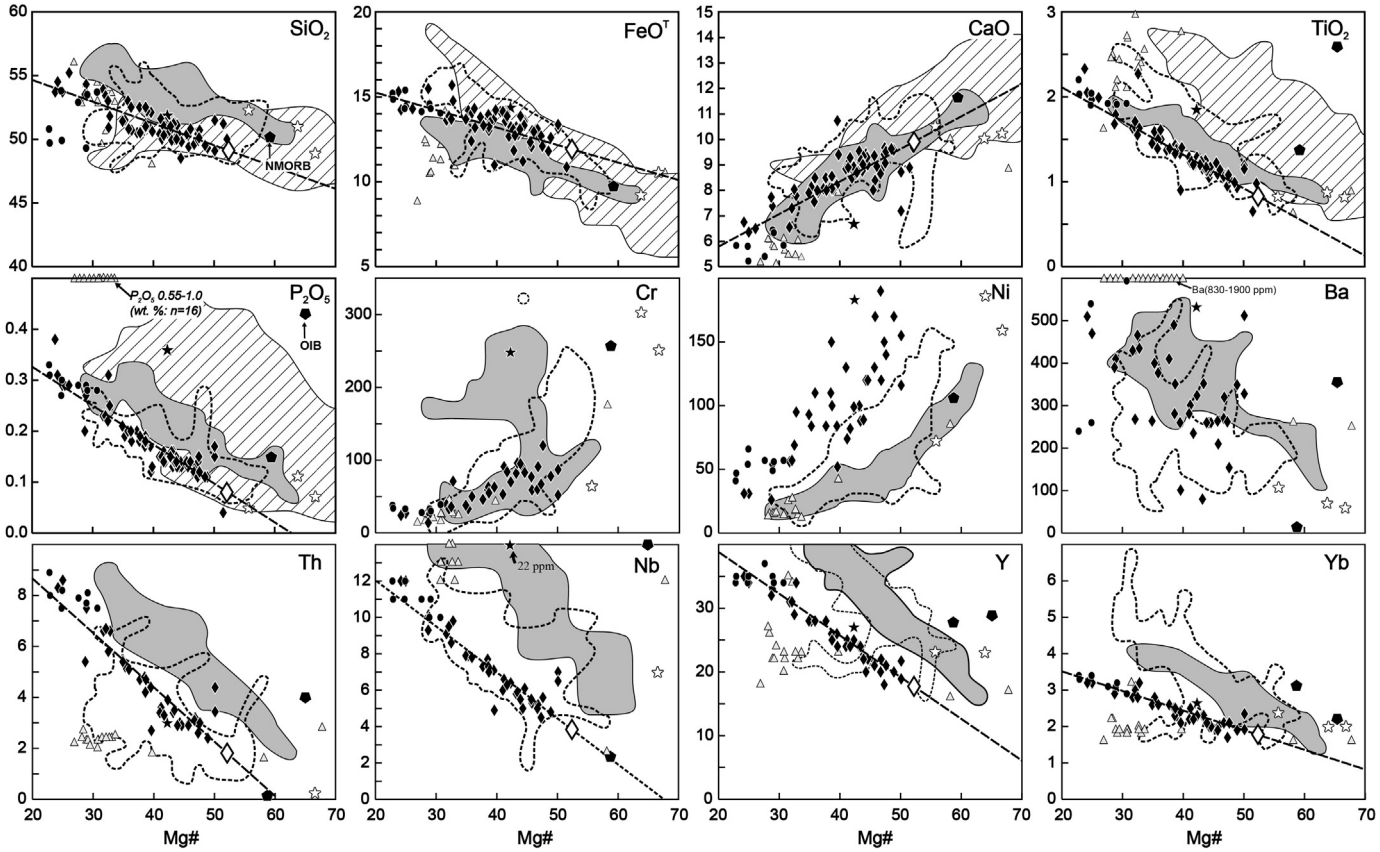


Fig. 8. Selected major and trace element data vs. Mg# for the Kaminak dykes and Spi basalts. Fields, symbols and references as in Fig. 6. Diagonally ruled field in major element diagrams is for mid ocean ridge basalts (N-MORB; Lehnert et al., 2000; http://www.petdb.org/). Filled pentagon is average N-MORB, filled upside down pentagon is OIB (Sun and McDonough, 1989). Large open diamond in major element diagrams is estimated most primitive Kaminak dyke composition determined by linear regression.

mantle plumes and the continental lithospheric mantle (CLM) in their origin. This extensive body of work has shown that many Phanerozoic continental flood basalt provinces contain at least two distinct end-member magma types that are referred to as high- and low-TiO_2 continental flood basalts. High-TiO_2 basalts are enriched in most REE, may exhibit weak HFSE troughs and have significantly less radiogenic Sr and more strongly radiogenic Nd relative to low-TiO_2 basalts. These features are characteristic of "plume-related" oceanic island basalts (OIB) derived via deep melting of the asthenosphere (Baker et al., 1996; Hart et al., 1989; Kieffer et al., 2004; McKenzie and O'Nions, 1995; Pearce, 2008; Presnall and Gudfinnsson, 2011; Stein and Goldstein, 1996). These OIB-like, high-TiO_2 basalts commonly vary to more depleted, N-MORB-like rocks. Thus, the proportion contributed from a depleted, N-MORB-type asthenospheric mantle reservoir may vary significantly both temporally and spatially within and between continental LIPs.

The origin of low-TiO_2 continental flood basalts is more contentious. Argument revolves around identifying the role of crustal vs. lithospheric mantle input into their primary magmas and indeed, whether the CLM contributes in any way to the origin of CFB. Many low-TiO_2 continental basalts have been argued to be the products of extensive assimilation-fractional crystallization (AFC) in the crust or continental lithosphere of deep, asthenosphere-derived (plume) primitive magmas (see discussion in Arndt and Christensen, 1992; Boily and Ludden, 1991; Dupuy and Dostal, 1984; Garfunkel, 2008; Thompson et al., 1983; Turner and Hawkesworth, 1995). Much of this work has emphasized that the primary, mantle-derived magmas of continental flood basalts are probably picritic, deep mantle, plume-derived magmas that pond at the Moho and undergo significant Ol + Cpx + Pl crystallization and potentially assimilate local lower crust (e.g., Cox, 1980).

Others have argued (e.g., Ellam and Cox, 1991) that, although the thermal anomaly and the majority of its derivative basaltic melts likely originate in the deep mantle and are thus permissive of a mantle plume origin, the non-radiogenic Nd, radiogenic Sr, LILE- and LREE-enriched but HFSE-depleted characteristics of some LIP basalts (e.g., many Gondwanan CFB) support their derivation through mixing of depleted asthenospheric, and LILE- and LREE-enriched CLM components. Ellam and Cox (1991) proposed that the lithospheric component comprised low-degree lamproitic partial melts of ancient metasomatized peridotite (cf., Foley, 1992; Mitchell and Bergman, 1991; Rock, 1991) that were scavenged from the CLM during upward passage of the deeper, asthenosphere-derived melts. However, the low TiO_2 abundances in such flood basalts are problematic, as typical lamproite has very high TiO_2 (2.64–8.12 wt.% TiO_2; Jacques et al., 1986). Examination of the whole-rock compositions of CFB led Gallagher and Hawkesworth (1992) to subsequently propose that even in the presence of a mantle plume, the majority of basaltic magma, particularly in situations where a thick CLM exists, may be derived wholly via dehydration melting of metasomatized CLM peridotite within the mechanical boundary layer at the base of the lithosphere. Turner and Hawkesworth (1995) further argued that the evolved major element compositions of low-TiO_2 CFB magmas may be generated through partial melting of a hydrated (metasomatized), previously melt depleted, continental lithospheric mantle peridotitic source at depths significantly shallower than the mechanical boundary layer.

The role of the CLM in LIP magmatism is, therefore, challenging to assess owing to our limited knowledge of its physical, thermal

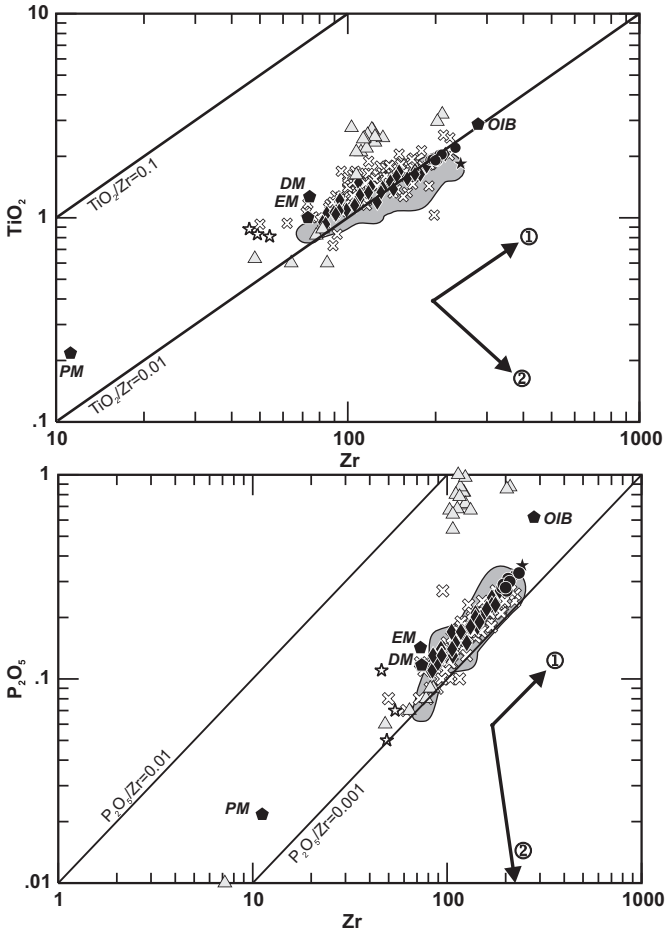


Fig. 9. (A) TiO_2 (wt.%) vs. Zr and (B) P_2O_5 (wt.%) vs. Zr (after Pearce and Norry, 1979) plots for the Kaminak dykes and Spi Group basalts. Fields, symbols and references as in Fig. 6 except type-2 Matachewan dykes are open crosses. Mantle components include: DM – depleted MORB, EM – enriched MORB, OIB – oceanic Island basalt, and PM – primitive mantle (Sun and McDonough, 1989). Arrow 1 represents 50% fractional crystallization of the assemblage 50% Pl + 45% Cpx + 5% Ol, and arrow 2 the assemblage 50% Pl + 40% Cpx + 5% Ol + 5% Mt, using sample DMA400-1967 as a source and applying the distribution coefficients of Rollinson (1993) and those from the GERM Partition Coefficient (Kd) Database (htttp://earthref.org/KDD/e:15/?&sort=mineral).

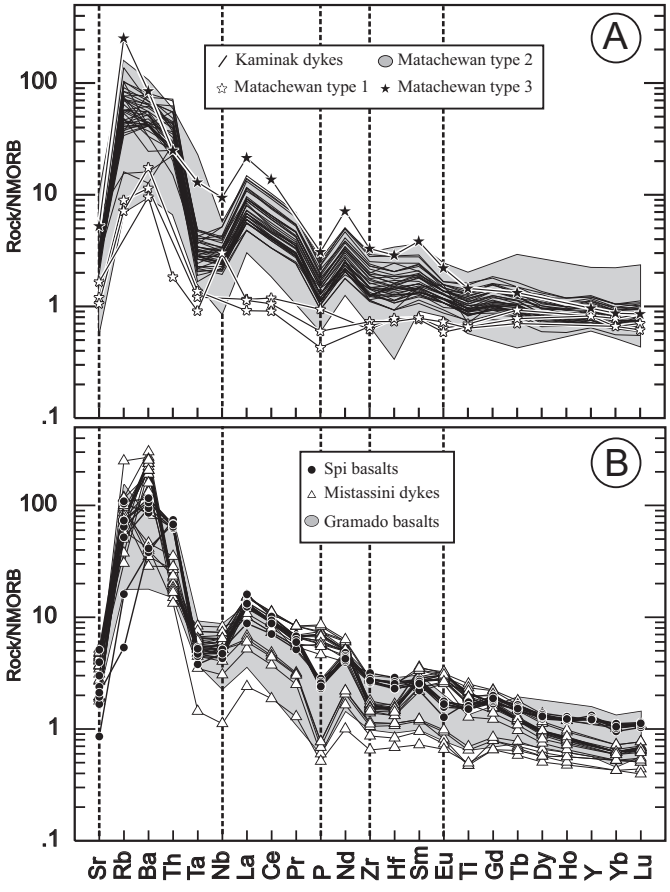


Fig. 10. (A) N-MORB-normalized multi-element (Sun and McDonough, 1989) plots for the Kaminak dykes compared to the Matachewan dykes and (B) N-MORB-normalized multi-element data for the Spi Group basalts, the low-TiO_2 Gramado basalts and the Mistassini dykes.

and chemical structure, data mainly derived via geophysical observations and direct sampling of rare mantle xenoliths entrained in alkali basalts and kimberlites. Because of the recent growth in publicly available data from the global diamond industry, it has become increasingly apparent, however, that the CLM beneath the majority of Archean cratons is heterogeneous, is commonly coarsely layered, and has been subjected to widespread and locally multiple metasomatic fluid infiltration episodes (Carbno and Canil, 2002; Downes, 2001; Griffin et al., 2008; Heaman et al., 2006). In light of these new observations, it is now more reasonable to assume that partial melting in the CLM may occur when the host peridotite has been previously fluxed by hydrous siliceous fluids, and moreover, would likely involve numerous distinct mantle source rocks ranging from dunite through harzburgite to lherzolite and pyroxenite. Partial melting of the lithospheric mantle therefore need not require lherzolitic sources and anhydrous conditions (e.g., Arndt and Christensen, 1992; Garfunkel, 2008). Abundant evidence now indicates that Archean CLM contains substantial volumes of metasomatized peridotites (*sensu lato*) that might contribute low-degree, LILE- and LREE-enriched, low-degree partial melts, either to more primitive, deeper asthenosphere-derived magmas, or to in situ melts of refertilized, variably melt-depleted, old CLM. The CLM source lithologies that might contribute such melts may be highly variable in their initial bulk composition prior to metasomatic modification, and, moreover, they may contain variable proportions of metasomatic veins (Foley, 1992). Metasomatic veins in the CLM source may themselves vary greatly in composition, from peridotite infiltrated by SiO_2–Al_2O_3-rich felsic melts through a range of fluids that may include variable abundances of H_2O, F, Cl, CH_4 or CO_2. Local differences in the compositions of the metasomatic veins and their host rocks will result in large compositional variations in the derivative low-degree partial melts, leading to initial melt compositions that may range from perhaps kimberlite and carbonatite through lamproite, kamafugite, minette and calc-alkaline lamprophyres.

6.1. Petrogenesis of the Kaminak dykes

The final petrogenetic process to influence the compositional variation of the Kaminak dykes appear to have occurred deep in the crust, and included fractional crystallization of Pl + Cpx ± Ol and perhaps assimilation of crust. The viability of an in situ, deep crustal fractional crystallization- assimilation model was tested using DMA400-1967, the most primitive of the Kaminak dyke samples. Regression lines of the co-linear major and incompatible trace element compositions (Fig. 8) project up Mg# (i.e., towards more primitive compositions) towards DMA400-1967, which has Mg# = 47.3 (MgO = 6.4), Cr = 67 ppm and Ni = 150 ppm. This composition approaches that of average N-MORB in terms of SiO_2, Al_2O_3, FeO_T and Na_2O, but has significantly elevated K_2O and much lower CaO, TiO_2 and P_2O_5.

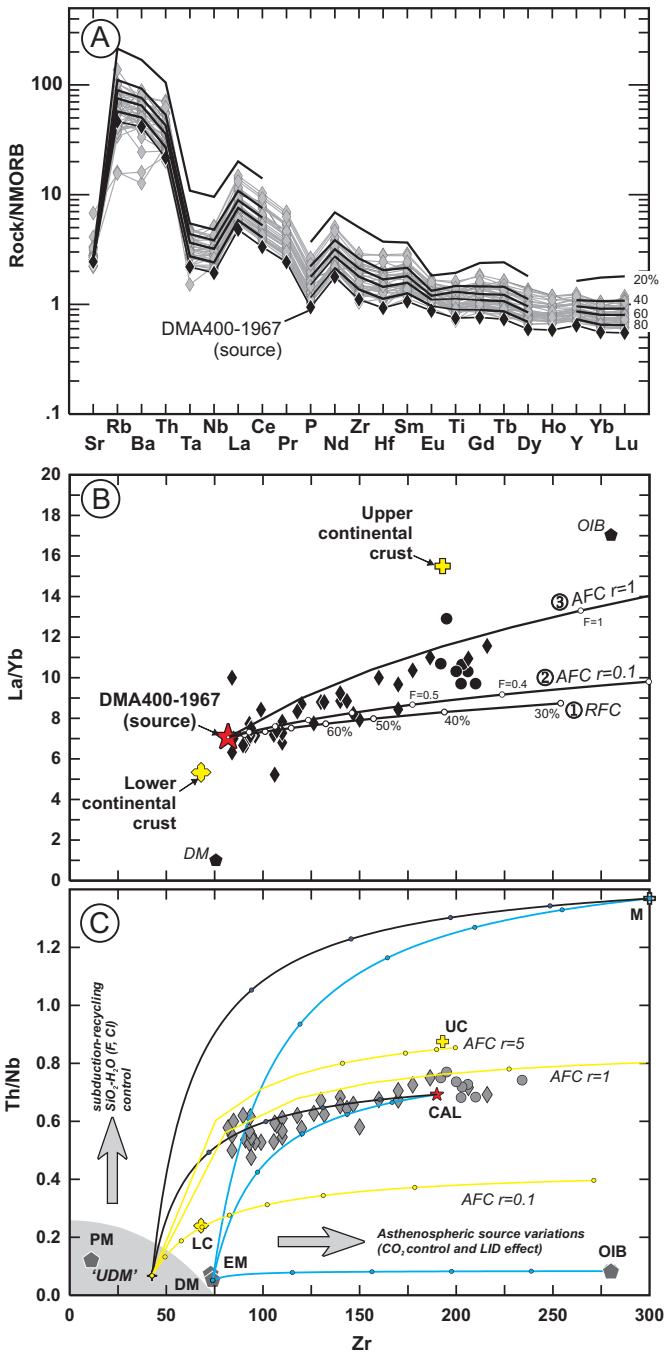


Fig. 11. (A) N-MORB-normalized multi-element plot showing the results for Rayleigh fractional crystallization (RFC) of Kaminak dyke sample DMA400-1967. All Kaminak dykes are shaded grey; calculated liquids are shown as heavy black lines with percent residual liquid remaining shown at right. Distribution coefficients are from Rollinson (1993) using an assemblage consisting of 50% Pl + 45% Cpx + 5% Ol; (B) plot of La/Yb vs. Zr showing modeled results (curve 1) of RFC of DMA400-1967 (assemblage given in A) with the percent residual liquid shown as white dots; curves 2 and 3 show combined AFC of upper continental crust (Rudnick and Gao, 2006) with a starting composition of DMA400-1967. *R*-values (ratio of assimilation rate to fractional crystallization rate) of $r = 0.1$ and 1 are shown. White dots represent fraction of remaining magma in 10% increments; (C) Th/Nb vs. Zr diagram illustrating the unrealistically high *r*-values required during assimilation of upper crust to produce their strongly elevated Th/Nb ratios, and that the Kaminak dykes were likely sourced from a metasomatically modified depleted mantle source containing low Zr and Nb. Mixing curves (Langmuir et al., 1978) between mantle components are shown in blue. Mixing of a primitive melt represented by a 50% depleted mantle (DM)–50% primitive mantle (PM) composition with either average calc-alkaline lamprophyre or minette readily explains the high Th/Nb and low Zr of the most primitive of the dykes. Plotted for comparison in B and C are: DM – depleted mantle; EM – enriched MORB; PM – primitive mantle: UDM – ultra depleted mantle; OIB – ocean island basalts (Sun and McDonough, 1989); lower crust (LC) and upper crust (UC) from Rudnick and Gao (2006); average granitoid crust of the Hearne craton (CHG; Sandeman et al., 2004a,b; Cousens et al., 2004); average compositions of minette (M) and calc-alkaline lamprophyre (CAL; Rock, 1991).

Calculated liquids for Rayleigh fractional crystallization of 50% Pl + 45% Cpx + 5% Ol from a parental magma having the composition of sample DMA400-1967 are shown in Fig. 11A. Basaltic distribution coefficients used are from Rollinson (1993). The modeled residual melts have multi-element patterns very similar to those of the Kaminak dykes, and for the more evolved dykes, range up to ~60% fractional crystallization of the source magma. The light-REE, Th and commonly Rb are, however, typically higher than the modeled results, particularly in the more evolved dykes. Examination of binary plots of incompatible trace element ratios (e.g., La/Yb vs. Zr; Fig. 11B), indicates that the range of dyke compositions cannot be easily produced; the residual liquids produced via fractional crystallization of the assemblage given above diverge significantly from the trends exhibited by the dykes (Fig. 11B). Assimilation fractional crystallization models using DMA400-1967 as the source magma and upper crust (UC) of Rudnick and Gao (2006) are also presented for *r*-values (r = ratio of assimilation rate vs. fractional crystallization rate) of 0.1 and 1, respectively. The majority of the Kaminak dykes fall on a trend that lies between these two calculated curves, suggesting that modest to extensive assimilation-fractional crystallization of evolved, upper crust may explain the trends. AFC of average lower continental crust cannot, however, produce the elevated La/Yb observed in the Kaminak dykes. Fig. 11C plots the Kaminak dykes in Th/Nb vs. Zr space and compares them to hypothetical asthenospheric end-member basalt types and primitive mantle (DM = depleted mantle; OIB = ocean Island basalt; PM = primitive mantle; Sun and McDonough, 1989), average calc-alkaline lamprophyre (CAL) and minette (M; Rock, 1991), and to upper and lower crust of Rudnick and Gao (2006). In order to produce the strongly elevated Th/Nb (~0.5) of the Kaminak dykes at correspondingly low Zr contents (80 ppm), *r*-values need be unrealistically high (~1–5). The primary Kaminak magmas had low (<75 ppm) Zr abundances (Fig. 11C), and thus their mantle source was depleted with respect to Zr and the other HFSE.

Four trace element ratio diagrams further help to constrain the origin of the Kaminak dykes. A modified Th/Ta vs. La/Nb diagram (after Condie, 1997) aids in evaluating the potential significance of possible petrochemical contributions from a number of recognized mantle, crustal and CLM end-members that might be involved in the genesis of basaltic suites (Fig. 12A). The Kaminak dykes have high Th/Ta ratios, lying well above those compositions defined by globally recognized, geochemically defined asthenospheric mantle components (Sun and McDonough, 1989). They therefore represent Proterozoic CFB that contain a continental lithospheric mantle, high Th/Ta component, as advocated by Condie (1997). Average upper continental crust (Rudnick and Gao, 2006) has a Th/Ta ratio similar to an average Kaminak dyke, however, an average of exposed Hearne craton granitoids (Cousens et al., 2004; Sandeman et al., 2004a,b) has Th/Ta lower than most of the dykes. Therefore, assimilation of normal, average Hearne granitoid crust from the present exposure level, by Kaminak mafic magmas cannot produce such elevated Th/Ta ratios.

A plot of Th/Yb vs. Ta/Yb (Pearce, 1982) underscores the elevated Th/Yb of the Kaminak dykes (Fig. 12B). Assimilation-fractional crystallization of Hearne craton granitoids can produce the range of Ta/Yb values for the Kaminak dykes, but cannot explain the notable variation in Th/Yb. Similarly, a plot of La/Sm vs. Th/Nb (Fig. 12C) emphasizes the high La/Sm ratios observed in the Kaminak dykes. Fractional crystallization and AFC processes can account for much of the variation in La/Sm, however, these cannot easily generate

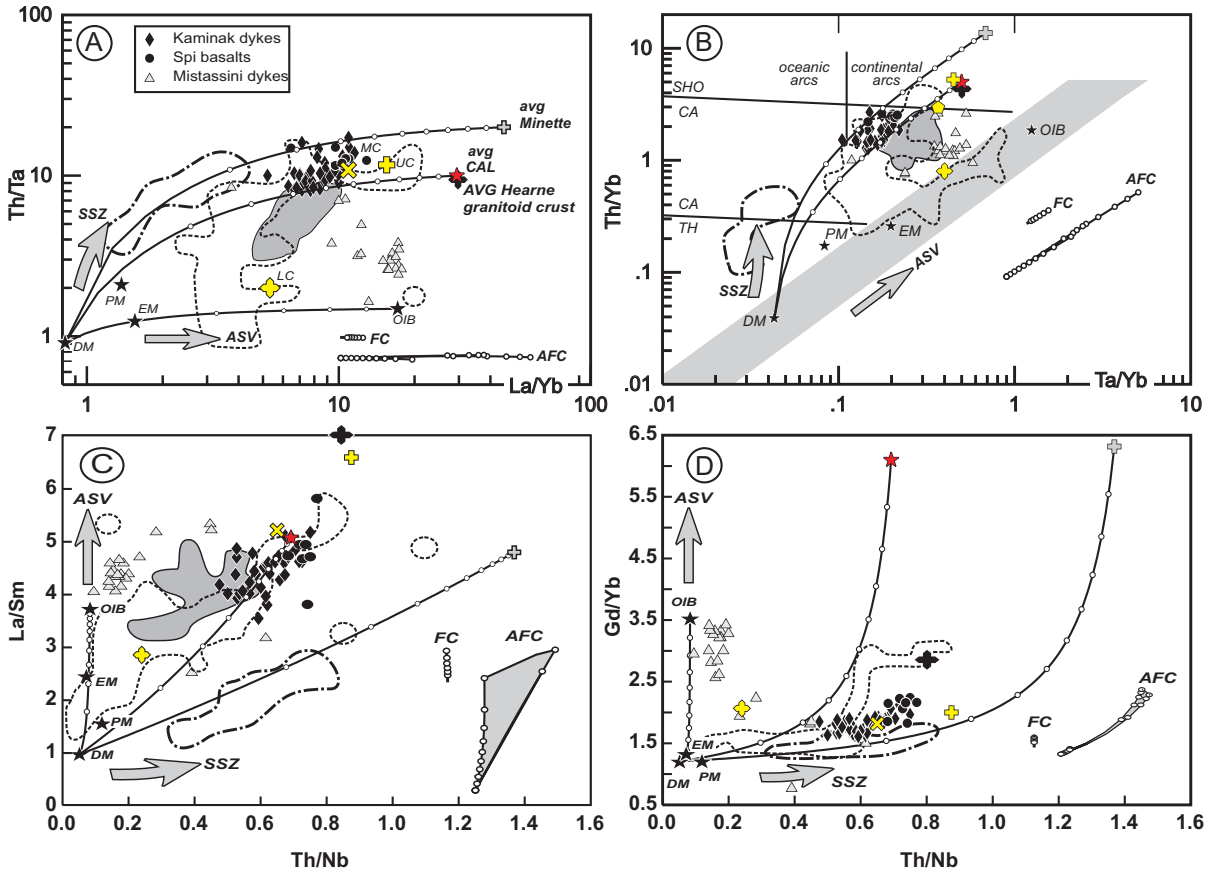


Fig. 12. Incompatible element plots for the Kaminak dykes and Spi basalts. (A) Th/Ta vs. La/Yb (adapted after Condie, 1997). (B) Th/Yb vs. Ta/Yb (adapted from Pearce, 1982). (C) La/Sm vs. Th/Nb. (D) Gd/Yb vs. Th/Nb. Fields, symbols and references as in Fig. 6. Abbreviations as in Fig. 11. Dash-dot field is for the ensimatic Mariana Arc (Elliott et al., 1997) showing supra-subduction zone effects. Curves labeled AFC represent trajectories and envelopes for assimilation fractional crystallization (DePaolo, 1981b) of Hearne craton average granitoid crust, applying *r*-values of 0.1, 1.0 and 5, and using sample DMA400-1967 as the parent and average Hearne craton granitoid as the assimilant. Dots represent *F*-values (*F* = mass assimilant/mass melt) from 0 to 5. Curves labeled FC represent Rayleigh fractional crystallization of the assemblage Pl + Cpx + Ol (50:45:5) from sample DMA400-1967. White dots represent % residual liquid remaining in gradations of 10%. Solid lines joining DM-OIB-minette and calc-alkaline lamprophyres are simple binary mixtures (Langmuir et al., 1978) of mantle end members. Shaded arrows show the general effects resulting from asthenospheric source variations (ASV) and suprasubduction zone enrichment (SSZ).

the large variation observed in the Th/Nb ratios. A plot of Gd/Yb vs. Th/Nb (Fig. 12D) illustrates the weak enrichment of the middle relative to heavy REE (HREE) as well as the strong Th/Nb enrichment exhibited by the Kaminak dykes. Fractional crystallization cannot account for the range in these ratios, although AFC may account for some of these variations. The collective major and trace element variations and abundances and the observations derived from the trace element ratio diagrams indicates that the Kaminak dykes appear to represent bulk mixtures of a depleted N-MORB-like mantle (DM) derived either from the CLM or deep asthenosphere, and an end-member having high LILE and LREE relative to the HFSE and HREE (e.g., Condie, 1997). The enriched component might be represented by the generally scarce, most fractionated and evolved Hearne craton granitoids, or perhaps a primary, incompatible element enriched, low-degree mantle melt component, here represented by average minette and calc-alkaline lamprophyre compositions (Rock, 1991). Minettes and calc-alkaline lamprophyres are significantly more common than lamproites (cf., Ellam and Cox, 1991), and have major, trace and radiogenic isotopic compositions more compatible with a viable CLM, low-melt-percent (1–5%) contaminant. In particular, minettes and calc-alkaline lamprophyres have elevated SiO_2 and K_2O, moderate FeO_T, MgO, CaO, Na_2O, and low bulk rock TiO_2 (1.10 and 1.30 wt.%, respectively), more suitable compositions than lamproite. As partial melting in the CLM progressively advances (10–20%), the primary magmas will gradually become less enriched in LILE and LREE, approaching more depleted, N-MORB-like melts that include progressively smaller proportions of these low-degree melt fractions. The Kaminak dykes therefore comprise one petrogenetic type that appear to have been derived via mixing large proportions of depleted mantle melts with an enriched lithospheric component.

Comparison (Fig. 13) of time-corrected Nd isotopic compositions for the Kaminak dykes to a number of other Paleoproterozoic continental mafic suites of the Hearne craton (Happotiyik Member of the Ameto Formation; Sandeman et al., 2003; Griffin gabbros; Aspler et al., 2002) and to Neoarchean magmatic rocks of the Hearne (Cousens et al., 2004; Sandeman et al., 2004a,b; Peterson et al., 2010) allow entertainment of four different petrogenetic scenarios. They might be derived from a CHUR-like mantle source with only minor subsequent isotopic modification, or originate: (1) by extensive partial melting of a silicic-melt-metasomatized and veined CLM; (2) from a deep sub-lithospheric source previously contaminated by early Neoarchean crust or; (3) by in situ crustal contamination of primary mantle-derived magmas. A mildly modified, isolated CHUR-like mantle source is unlikely based on modern ideas of a dynamic and convecting asthenosphere (e.g., van Keken et al., 2002), whereas the other three proposals are more plausible hypotheses. The dykes exhibit very poor correlations between $^{143}Nd/^{144}Nd_t$ and other important petrochemical indices. SiO_2 varies by ~5 wt.% among samples analyzed for Nd isotopes,

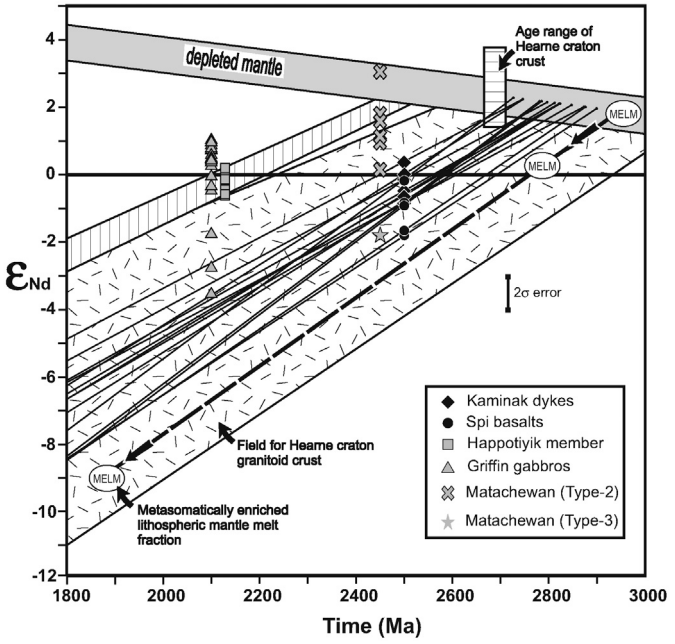


Fig. 13. ε_{Nd_t} versus time for 9 Kaminak dykes and 5 Spi basalts (Sandeman and Ryan, 2008), compared to other Paleoproterozoic basaltic rocks of the Hearne craton; the 2111 Ma Griffin gabbros (Aspler et al., 2002), and the undated Happotiyik Member of the Ameto Formation (Sandeman et al., 2003). Heavy solid lines are Nd isotopic evolution lines for the Kaminak dykes. Vertically ruled field encompasses all data for the Happotiyik Member. Hearne craton granitoid crust is from Sandeman et al. (2004a,b), Cousens et al. (2004) and Peterson et al. (2010). The depleted mantle curve is after DePaolo (1981a). MELM represents hypothetical metasomatized Neoarchean CLM and its isotopic evolution through time.

although corresponding ε_{Nd_t} values for the most SiO_2-poor and SiO_2-rich samples overlap, within error. Similarly the most MgO-rich and -poor specimens also have overlapping ε_{Nd_t} values. Other critical trace element ratios (Th/Nb, La/Yb, La/Sm, Th/Yb) for the Kaminak dykes also do not show systematic variation vs. ε_{Nd_t} values. These points indicate that significant assimilation of evolved, siliceous crustal material cannot fully explain the lithogeochemical and Nd isotopic data. The weak variations in the Nd data therefore appear to best support an origin for the Kaminak dyke primary magmas via lithospheric contamination in the mantle. If the deep crustal magmas chambers feeding the Kaminak dykes were hosted in Hearne crust, then Nd isotopic data alone cannot distinguish Hearne crust from contemporaneous CLM.

A prominent feature of the Kaminak dykes is that none appear to be derived exclusively from depleted mantle. As both Sm and Nd are LREE, addition of only a small proportion of LILE- and LREE-enriched material (crust or mantle) to a predominantly depleted asthenospheric melt is all that would be required to exert strong control on the REE budget of the derivative magma. Although this proposal was not quantitatively modeled, at $t = 2500$ Ma, a mixture of DM ($\varepsilon_{Nd_{2.5\,Ga}} = +2.56$) with a minor but likely variable (~1–5 vol.%) contribution of lamprophyric melt from metasomatically enriched CLM (CLM $\varepsilon_{Nd_{2.5\,Ga}} = -3.0$; Fig. 13) would satisfactorily explain the CHUR-like character of the data. As depicted (Fig. 13), a metasomatically veined lithospheric mantle source (MELM), established in the Hearne lithospheric mantle at ca. 2.7 Ga and comparable in trace element and Nd isotopic composition to average calc-alkaline lamprophyre, would be an appropriate addition to the CLM at the time in order to facilitate the ca. 2.5 Ga production of low-degree melt fractions during Kaminak magmatism. Although the metasomatic veins likely contributed to Kaminak dyke magmatism, it is unknown whether their source was entirely from the CLM, or the source comprised a mixture of upwelling, DM magmas with small volumes of magma from re-fertilized CLM. In the absence of picritic or high-Ti OIB-like magmas of unequivocal deep (mantle-plume?) origin, the cause of the melting and basaltic magmatism was likely not a large-scale mantle plume. Instead, melting is inferred to have occurred in the CLM in response to a passive thermochemical anomaly in the asthenosphere (e.g., Anderson, 2005) associated with lithospheric transtension of the Hearne craton during the early Paleoproterozoic (ca. 2498 Ma). Asthenospheric upwelling at that time may have been focused by topographic irregularities in the base of the lithosphere resulting from Neoarchean cratonization processes.

Collectively the major, incompatible trace element and Nd isotopic variations of the Kaminak dykes and Spi basalts are most compatible with their primary mantle-derived magmas containing depleted N-MORB-like and LILE- and LREE-enriched CLM components. In this way they are comparable to the low-TiO_2 Mesozoic Gramado basalts of the Paraná LIP, South America (Garland et al., 1996; Peate and Hawkesworth, 1996; Turner and Hawkesworth, 1995). On the basis of these broad similarities and the geochemical arguments presented above, we infer that the Kaminak dykes were similarly derived via partial melting of a previously melt-depleted CLM, which stabilized in the Neoarchean and was infiltrated at that time by subduction-related, predominantly juvenile, hydrous silicate melts that contained a minor component of older, fractionated material.

6.2. Regional implications

Based on the work of the western Churchill NATMAP project, it is now recognized that much of the Hearne craton of the western Churchill Province represents juvenile, Neoarchean ensimatic crust that was formed and stabilized during the interval 2715–2660 Ma (Cousens et al., 2004; Davis et al., 2004; Hanmer et al., 2004; Sandeman et al., 2004a,b). Many of the Neoarchean igneous rocks of the Hearne have lithogeochemical and isotopic characteristics, such as prominent negative HFSE anomalies and ε_{Nd_t} values slightly lower than DM, indicating their derivation from a juvenile, silicic fluid-modified asthenospheric mantle. The amalgamation of the Hearne craton is therefore considered to have occurred in conjunction with subduction of juvenile Neoarchean oceanic crust that was accompanied by metasomatic infiltration of the Hearne CLM by LILE- and LREE-enriched fluids (e.g., Cousens et al., 2001, 2004; Sandeman et al., 2004a,b). This proposal is further supported by three observations: (1) T_{DM} ages for some felsic granitoid and volcanic rocks of the Hearne are modestly older than their crust formation ages ($T_{DM's}$ = 2.73–3.10 Ga; mean = 2.76 Ga); (2) Neoarchean (2682 ± 2 Ma; Davis et al., 2004) monzonitic rocks of the Carr Lake pluton have compositions (Sandeman et al., 2004a) comparable to high-Mg monzodiorite or Sanukitoid suite rocks (Stern and Hanson, 1991); and (3) a Neoarchean (2659 ± 5 Ma), late-cratonization, carbonatite–ijolite–jacupirangite complex is present in the area (Cavell et al., 1992; Davidson, 1970).

Geochemical and isotopic characteristics suggesting magma derivation from, and/or interaction with, metasomatized CLM are observed in the lithogeochemical and Nd isotopic compositions of many of the Paleoproterozoic, intraplate mafic rocks that occur in the Hearne craton (Aspler et al., 2002; Sandeman et al., 2003; Sandeman, 2005, 2007; Sandeman and Ryan, 2008). These compositional features reflect ~250 m.y. of mantle-derived mafic magmatism that may have sampled such a CLM source. For example, the ca. 2.2 Ga Happotiyik Member of the Ameto Formation also shows LILE- and LREE-enrichment and time corrected Nd isotopic compositions that range to CHUR-like values (Fig. 13; Sandeman et al., 2003). Similarly, the Griffin sills of the Hurwitz Group, dated at 2.111 Ga, exhibit LILE- and LREE-enrichment and have time corrected Nd isotopic compositions that exhibit a broad range from +1

to −4 (Aspler et al., 2002), overlapping those of the Kaminak dykes and the Happotiyik Member (Fig. 13; Sandeman et al., 2003). In light of these data, the compositional features of many of the Hearne craton's Paleoproterozoic mafic rocks suggest that they were likely derived, at least in part, from a continental lithospheric mantle source that had experienced modest LILE- and LREE-enrichment during formation and amalgamation of the Neoarchean Hearne craton (Sandeman et al., 2000; Cousens et al., 2001). These observations imply that such a veined and metasomatically modified mantle source likely existed beneath the Hearne craton during much of the Paleoproterozoic, and that the CLM beneath the Hearne craton was a viable source for all of these CFB suites.

6.3. Comparison with the Matachewan and Mistassini swarms

Bleeker (2004), Bleeker and Ernst (2006) and Ernst and Bleeker (2010) correlated the Kaminak dykes with the Matachewan dykes of the Superior Province (Boily and Ludden, 1991; Nelson et al., 1990; Phinney and Halls, 2001). This correlation is based on broad similarity in age and paleolatitude, common occurrence of large plagioclase phenocrysts/glomerocrysts in dykes of both swarms and the similarity in N–S dyke swarm orientation. These swarms were therefore inferred to form a significant lithotectonic pin for Archean continental reconstruction, placing the northern margin of the Hearne craton along the southern margin of the Superior craton at ca. 2.5 Ga. Based on similar, ca. 2.5 Ga crystallization ages for the Mistassini and Ptarmigan swarms, Ernst and Bleeker (2010) proposed that these were also Kaminak dyke correlatives. However, comparison of a number of salient lithogeochemical characteristics of the Kaminak, Matachewan, and Mistassini dyke swarms do not support their direct correlation. Firstly, the Kaminak dykes represent tholeiitic magmas that crystallized under high-P, lower crustal conditions, whereas many of the Matachewan dykes are characterized by normative compositions indicating they have evolved via fractionation of Pl + Cpx ± Ol under low-P, upper crustal conditions. The Mistassini dykes appear to have crystallized under lower- to middle-crustal conditions, but range to much more primitive compositions (e.g., Mg#'s 26.8–85.3; Cr = 13–3280 ppm; Ni = 13–895 ppm; Ernst and Buchan, 2010) than the Kaminak dykes.

Secondly, Matachewan dykes show greater compositional variation than the Kaminak dykes, and, moreover, are represented by at least three and possibly up to six petrochemically distinct subswarms (Phinney and Halls, 2001). The majority of the Matachewan dykes comprise a group that exhibits intermediate REE and incompatible trace element abundances, is variably LREE-enriched [$(La/Yb)_n$ = 0.69–14.5; mean = 2.99, n = 39], and shows similar variable Nb, Ti, and minor Eu troughs (Fig. 10). This group is represented by the type-2 dykes reported by Phinney and Halls (2001), and records extended trace element patterns and abundances somewhat similar to those for the Kaminak dykes. The major significant distinctions are that the Kaminak dykes have greater LILE- and LREE-enrichment and depleted heavy-REE (HREE) and Y abundances relative to the Type-2 Matachewan dykes. Moreover, the single petrogenetic group of Kaminak dykes does not match well with any of the 6 subtypes recognized by Phinney and Halls (2001). Mistassini dykes form at least two subgroups, both of which are petrochemically distinct from the Kaminak and Matachewan dykes (Fig. 10). Most Mistassini dykes have very high TiO_2 and P_2O_5 and have trace element ratios that indicate significant contributions from an asthenospheric, OIB-like source. Such a signature is minor or absent in the Kaminak and the Type-2 Matachewan dykes.

Thirdly, the more radiogenic $^{143}Nd/^{144}Nd$ isotopic compositions of the Kaminak dykes and the Spi Group basalts relative to the Type-2 Matachewan dykes (Boily and Ludden, 1991) indicate that the ca. 2.5 Ga Hearne diabase magmas contain a greater proportion of an evolved, Neoarchean lithospheric component (Fig. 13). The single Type-3 Matachewan dyke has a $^{143}Nd/^{144}Nd$ isotopic composition that overlaps with 2 of the Spi Group basalts, but this sample is characterized by greater LREE-enrichment and significantly elevated HFSE abundances relative to the Kaminak dykes.

Finally, the 2497.6 ± 1.1 Ma U–Pb baddeleyite date obtained in this study for Kaminak dyke PHA97-N87 is older than any known Matachewan dyke emplacement ages in the southern Superior Province. The range of U–Pb ages previously reported for the Matachewan dykes is $2473^{+16}/_{-9}$ Ma (Heaman, 1997) to 2459 ± 5 Ma (Halls et al., 2005). These ages are distinctly younger and do not overlap the robust age of the Kaminak dyke reported here. The Mistassini swarm has been dated at 2515 ± 3 Ma (Hamilton, 2009) and similarly does not overlap, within error, the age determination for the Kaminak swarm. Although there is likely to be at least a small range of emplacement ages for Kaminak dykes, as observed for a few other Proterozoic dyke swarms (e.g., 2100 Ma Marathon dyke swarm, Halls et al., 2008), at present there is no indication that Kaminak, Matachewan and Mistassini dyke emplacement was synchronous.

6.4. Implications for continental reconstructions and assembly of the western Canadian Shield

Cratons such as the Archean Superior craton, have granite–greenstone and structural belts that are truncated at their current margins, indicating that they were formerly parts of larger continental masses (e.g., Bleeker, 2003; Williams et al., 1991). The architecture of these larger 'continents' is a matter of continuing debate. Williams et al. (1991) proposed that the North American Archean cratons were once part of a single landmass they called 'Kenorland', and more recent reconstructions (e.g., Bleeker, 2003) have proposed multiple Archean landmasses, such as 'Sclavia' and 'Superia', based on geological similarities in different crustal blocks. In the most recent Neoarchean reconstruction (Bleeker, 2004; Bleeker and Ernst, 2006; Ernst and Bleeker, 2010), 'Superia' is an amalgam consisting of at least four Archean crustal blocks; three currently located within North America (Superior, Wyoming, and Hearne) and one from the Baltic Shield (Karelia). The primary reasons for locating the northern margin of the Hearne craton along the southern margin of the Superior Province at the end of the Neoarchean are similarity in formation ages and structural trends of granite–greenstone belts, ca. 2.45 Ga dykes with coeval ages (Heaman, 1994, 1997), physical appearance, paleolatitude, and ~N–S orientation (i.e. Matachewan and Kaminak swarms), as well as ca. 2.11 Ga mafic magmatism (i.e. Griffin sills, Marathon dykes). In the original reconstruction of Bleeker (2004), these dyke swarms were emplaced into a single continent during rifting and eventual breakup of Superia at about 2.45 Ga. In a more recent reconstruction (Bleeker and Ernst, 2006; Ernst and Bleeker, 2010), breakup of this region of Superia was interpreted to have occurred at ~2.1 Ga in this region of Superia. The latter model implies that the Matachewan, Mistassini and Kaminak swarms are LIP basalts associated only with extension of the lithosphere, and did not promote cratonic separation and the formation of an intervening ocean basin. Hence, these were generated by passive mantle upwelling and lithospheric extension as outlined by Anderson (2005).

An important question in evaluating the veracity of proposed reconstructions is, therefore, whether any of the LIPs used as pins are part of more localized extension (no resulting lithospheric separation) vs. regional, catastrophic rifting events. The former are not, whereas the latter may represent unique cratonic reconstruction pins. There is a strong case that the ~2.4–2.5 Ga mafic magmatism was a widespread event that is represented on many crustal fragments but did not result in fragmentation of the

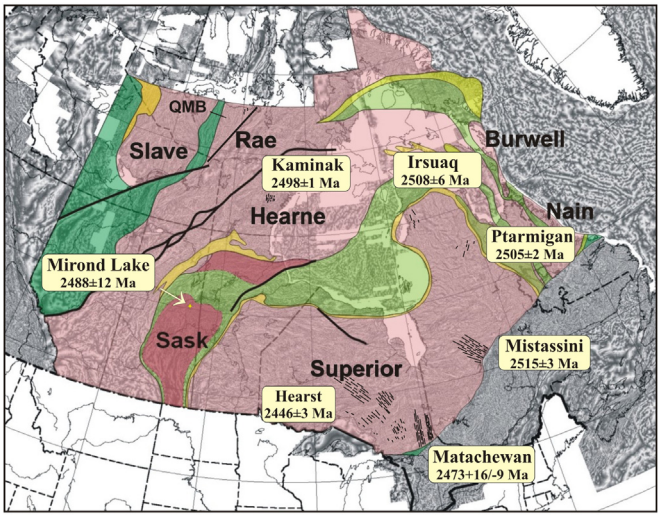


Fig. 14. Simplified geological map of central Canada showing cratonic blocks and the locations, spatial extents and U–Pb baddeleyite ages for ca. 2500 Ma dyke swarms (Heaman, 1994, 1997; Buchan et al., 1998; Halls et al., 2005; Maurice et al., 2009).

supercontinent. Of particular importance here is that there are numerous isolated, ca. 2.5 Ga mafic dyke swarms located along the present-day southern margin of the Superior Province (Fig. 14). These include the 2450–2490 Ma Matachewan (Halls et al., 2008), 2510 Ma Mistassini (Hamilton, 2009), 2500 Ma Ptarmigan (Buchan et al., 1998), and 2510 Ma Irsuaq (Maurice et al., 2009) swarms. This ~2.5 Ga circum-Superior magmatic event might be a harbinger to subsequent circum-Superior mafic magmatism; for example at 1.88 Ga (Heaman et al., 2009). Therefore, based only on similar 2.5 Ga dyke dates in the Superior and Hearne cratons, it is unclear whether the Kaminak dykes truly match with any ~2.5 Ga Superior Province dyke swarms. The geochemistry presented here indicates the Kaminak dykes represent a homogeneous, single petrogenetic suite, and are not an unequivocal petrochemical match with any of the distinct sub-types of the Matachewan or Mistassini swarms. Moreover, the available geochronological data for emplacement of the Kaminak, Matachewan and Mistassini dyke swarms do not overlap. Therefore our data do not provide unequivocal support for direct correlation of these swarms, and hence an unambiguous ca. 2.5 Ga, northern margin Hearne–southern margin Superior craton reconstruction. An alternative scenario is that the proto-Superior craton (Superia?) included some or all of the Hearne craton along the NW margin of the current Superior Province in the Neoarchean, and that crustal extension occurred at ca. 2.5 Ga, leading to the emplacement of the Spi basalts and Kaminak dykes, but not continental breakup. Subsequent rifting and breakup of Superia at ca. 2.1 Ga in this region was responsible for creating a continental fragment of Superia, now part of the Hearne craton, on the west side of the proposed Manikewan Ocean (Stauffer, 1984). Remnants of this 2.1 Ga rifting episode are preserved in the Hearne craton (2.111 Ga Griffin sills; Heaman and LeCheminant, 1993; Happotiyik Member, Hurwitz Group, Sandeman et al., 2003), Wollaston Domain (>2.075 Ga Courtney Lake rift sequence; Ansdell et al., 2000), and the NW Superior Province margin, including 2.1 Ga mafic dykes in the Split Lake Block (Heaman and Corkery, 1996) and the Pikwitonei Granulite Domain (Halls and Heaman, 2000). This hypothesis is testable. If correct, there should be 2.5 Ga mafic dykes located along the NW margin of the Superior Province that represent the conjugate margin extension of the Kaminak dyke swarm.

7. Conclusions

A two fraction, multi-grain, U–Pb baddeleyite age for one Kaminak dyke (PHA97-N87) yielded a weighted mean $^{207}Pb/^{206}Pb$ date of 2497.6 ± 1.1 Ma (MSWD = 2.7; <1% discordant) that is interpreted as the time of baddeleyite crystallization in this sample, and a relatively accurate estimate for the time of the emplacement of this Kaminak dyke. This result is significantly older than, and does not overlap, within error, available ages for the Matachewan swarm of the Superior Province. Similarly, the Mistassini dykes of the Superior Province are older than and do not overlap, within error, this new age for the Kaminak dykes.

Robust lithogeochemical and Nd isotopic data indicate that the Kaminak, Matachewan and Mistassini dykes form distinct petrogenetic groups, although they exhibit some broad continental flood basalt similarities. The range in observed compositions cannot be easily attributed to fractionation or assimilation-fractional crystallization of common primary mantle-derived magmas, but must result from contributions from distinct mantle and/or crustal sources. The data indicate that the Kaminak dykes (and Spi Group basalts) form a suite of olivine- and quartz-normative, subalkaline, tholeiitic gabbroic and basaltic rocks that have homogeneous lithogeochemical compositions. They constitute a single petrogenetic group which formed via deep crustal fractional crystallization (and perhaps AFC) of the assemblage Pl + Cpx ± Ol, without magnetite or apatite fractionation. Low-TiO_2 and HFSE abundances approaching or lower than N-MORB compositions, particularly for the most primitive dykes, along with enrichment of LILE and LREE relative to the HFSE, are all features characteristic of low-TiO_2 continental tholeiites comparable to those found in Phanerozoic continental LIPs. The Kaminak dykes have homogeneous, CHUR-like Nd isotopic compositions and depleted mantle model ages that overlap extensively with those of Hearne craton's juvenile Archean crust. Comparable data for the Matachewan swarm indicate that the Kaminak dykes contain a greater proportion of a LREE-enriched, older lithospheric component. Collectively the data presented herein do not support direct correlation of the Kaminak dykes with the Matachewan and Mistassini swarms, and hence casts doubt on the model of a ca. 2.5 Ga connection between the northern Hearne and southern Superior margins (Bleeker and Ernst, 2006; Ernst and Bleeker, 2010). The proposed linkage of the Hearne and Superior cratons prior to Nuna supercontinent assembly requires further geological, geochronological, lithogeochemical and radiogenic isotopic constraints.

Acknowledgements

Early fieldwork and publications by Tony Davidson along with his archival Kaminak dyke samples were instrumental in the construction and completion of this manuscript. We are indebted to the technical and analytical staff of the Geological Survey of Canada and the Department of Earth Sciences of McGill and Memorial Universities for their laboratory support. Polar Continental Shelf Project, M & T Enterprises of Rankin Inlet and Huisson Aviation provided invaluable logistical support in the field. This research was completed as part of the multi-disciplinary Western Churchill NATMAP project, funded through the Geological Survey of Canada and NATMAP. The U–Pb geochronology research was supported through an NSERC Discovery grant and Major Resources Support grant to LMH. Rob Berman, Wouter Bleeker and two anonymous referees significantly improved the manuscript.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.precamres.2012.06.002.

References

Anderson, D.L., 2005. Scoring hotspots: the plume and plate paradigms. Geological Society of America Special Paper 388, 31–54.

Anderson, D.L., Tanimoto, T., Zhang, Y., 1992. Plate tectonics and hotspots: the third dimension. Science 256, 1645–1651.

Ansdell, K.M., MacNeil, A., Delaney, G.D., Hamilton, M.A., 2000. Rifting and development of the Hearne craton passive margin: age constraint from the Cook Lake area, Wollaston Domain, Trans-Hudson Orogen, Saskatchewan. In: Geological Association of Canada, GeoCanada 2000 Extended Abstract, vol. 777, p. 4.

Arndt, N.T., Christensen, U., 1992. Role of lithospheric mantle in continental volcanism: thermal and geochemical constraints. Journal of Geophysical Research 97, 10967–10981.

Aspler, L.B., Chiarenzelli, J.R., 1996. Relationship between the Montgomery Lake and Hurwitz Groups, and stratigraphic revision of the lower Hurwitz Group, District of Keewatin. Canadian Journal of Earth Sciences 33, 1243–1255.

Aspler, L.B., Chiarenzelli, J.R., 1997. Initiation of ca. 2.45–2.1 Ga intracratonic basin sedimentation of the Hurwitz Group, Keewatin hinterland, Northwest Territories, Canada. Precambrian Research 81, 265–297.

Aspler, L.B., Armitage, A.E., Ryan, J.J., Hauseux, M., Surmacz, S., Harvey, B.J.A., 2000. Precambrian geology, Victory and Mackenzie lakes, Nunavut, and significance of Mackenzie Lake metasediments, Paleoproterozoic Hurwitz Group. Geological Survey of Canada Paper 2000-C10, 10 p. (online; http://www.nrcan.gc.ca/gsc/bookstore).

Aspler, L.B., Cousens, B.L., Chiarenzelli, J.R., 2002. Long-distance intracratonic transport of mafic magmas during opening of the Manikewan ocean (Trans-Hudson orogen): Griffin gabbro sills (2111 Ma), Hurwitz Basin, Nunavut, Canada. Precambrian Research 117, 269–294.

Baker, J.A., Thirlwall, M.F., Menzies, M.A., 1996. Sr–Nd–Pb isotopic and trace element evidence for crustal contamination of plume-derived flood basalts: oligocene flood volcanism in western Yemen. Geochimica et Cosmochimica Acta 60, 2559–2581.

Baragar, W.R.A., Ernst, R.E., Hulbert, L., Peterson, T., 1996. Longitudinal petrochemical variation in the Mackenzie dyke swarm, northwestern Canadian Shield. Journal of Petrology 37, 317–359.

Beavon, R.V., 1976. Early Aphebian basaltic volcanism in the southern District of Keewatin, Northwest Territories. Canadian Journal of Earth Sciences 13, 1003–1005.

Bell, R.T., 1968. Preliminary Notes on the Proterozoic Hurwitz Group, Tavani (55 K) and Kaminak Lake (55L) Areas, District of Keewatin. Geological Survey of Canada, Paper 68-36, 17 p.

Bell, R.T., 1971. Geology of Henik Lakes (East Half) and Ferguson Lake (East Half) Map-areas, District of Keewatin. Geological Survey of Canada, Paper 70-61, 31 p.

Berman, R.G., Davis, W.J., Ryan, J.J., Tella, S., Brown, N., 2002. In-situ sensitive high resolution ion microprobe (SHRIMP) U/Pb geochronology of Barrovian facies-series metasedimentary rocks in the Happy Lake and Josephine River supracrustal belts: implications for the Paleoproterozoic architecture of the northern Hearne craton, Nunavut. In: Radiogenic Age and Isotope Studies; Report 15. Geological Survey of Canada, Current Research, 2002-F4, 14p.

Berman, R.G., Davis, W.J., Pehrsson, S., 2007. The collisional Snowbird tectonic zone resurrected: growth of Laurentia during the 1.9 Ga accretionary phase of the Trans-Hudson orogeny. Geology 35, 911–914.

Bleeker, W., 2003. The Archean record: a puzzle in ca. 35 pieces. Lithos 71, 99–134.

Bleeker, W., 2004. Taking the pulse of planet Earth: a proposal for a new multidisciplinary flagship project in Canadian solid Earth Sciences. Geoscience Canada 31, 179–190.

Bleeker, W., Ernst, R., 2006. Short-lived mantle generated magmatic events and their dyke swarms: the key unlocking Earth's paleogeographic record back to 2.6 Ga. In: Hanski, E., Mertanen, S., Rämö, T., Vuollo, J. (Eds.), Dyke Swarms—Time Markers of Crustal Evolution. A.A. Balkema Publishers, Rotterdam.

Boily, M., Ludden, J.N., 1991. Trace-element and Nd isotopic variations in Early Proterozoic dyke swarms emplaced in the vicinity of the Kapuskasing structural zone: enriched mantle or assimilation and fractional crystallization (AFC) process? Canadian Journal of Earth Sciences 28, 26–36.

Bondre, N.R., Hart, W.K., Sheth, H.C., 2006. Geology and geochemistry of the Sangamner mafic dyke swarm, western Deccan volcanic province, India: implications for regional stratigraphy. Journal of Geology 114, 155–170.

Buchan, K.L., Mortensen, J.K., Card, K.D., Percival, J.A., 1998. Paleomagnetism and U–Pb geochronology of diabase dyke swarms of Minto block, Superior Province, Quebec, Canada. Canadian Journal of Earth Sciences 35, 1054–1069.

Buchan, K.L., Goutier, J., Hamilton, M.A., Ernst, R.E., Matthews, W.A., 2007. Paleomagnetism, U–Pb geochronology and geochemistry of Lac Esprit and other dyke swarms, James Bay area, Quebec, and implications for Paleoproterozoic deformation of the Superior Province. Canadian Journal of Earth Sciences 44, 643–664.

Cadman, A.C., Heaman, L., Tarney, J., Wardle, R., Krogh, T.E., 1993. U–Pb geochronology and geochemical variation within two Proterozoic mafic dyke swarms, Labrador. Canadian Journal of Earth Sciences 30, 1490–1504.

Cadman, A.C., Tarney, J., Hamilton, M.A., 1997. Petrogenetic relationship between Paleoproterozoic tholeiitic dykes and associated high-Mg noritic dykes, Labrador, Canada. Precambrian Research 82, 63–84.

Carbno, G.B., Canil, D., 2002. Mantle structure beneath the SW Slave craton, Canada: constrints from garnet geochemistry in the Drybones Bay kimberlite. Journal of Petrology 43, 129–142.

Carpenter, R.L., 2003. Relative and absolute timing of supracrustal deposition, tectonothermal activity and gold mineralization, West Meliadine region, Rankin Inlet greenstone belt, Nunavut, Canada. Unpublished Ph.D. Thesis. University of Western Ontario, London, Ontario, Canada, 386 p.

Cavell, P.A., Wijbrans, J.R., Baadsgaard, H., 1992. Archean magmatism in the Kaminak Lake area, District of Keewatin, Northwest Territories: ages of the carbonatite-bearing alkaline complex and some host granitoid rocks. Canadian Journal of Earth Sciences 29, 896–908.

Christie, K.W., Davidson, A., Fahrig, W.F., 1975. The paleomagnetism of Kaminak dikes: no evidence of significant Hudsonian plate motion. Canadian Journal of Earth Sciences 12, 2048–2064.

Coffin, M.F., Eldholm, O., 1994. Large Igneous Provinces: crustal structure, dimensions, and external consequences. Reviews of Geophysics 32, 1–36.

Condie, K., 1997. Sources of Proterozoic mafic dyke swarms: constraints from Th/Ta and La/Yb ratios. Precambrian Research 81, 3–14.

Condie, K.C., Bobrow, D.J., Card, K.D., 1987. Geochemistry of Precambrian Mafic dykes from the Southern Superior Province of the Canadian Shield. In: Halls, H.C., Fahrig, W.F. (Eds.), Mafic Dyke Swarms, vol. 34. Geological Association of Canada Special Paper, pp. 95–108.

Cousens, B.L., Aspler, L.B., Chiarenzelli, J.R., Donaldson, J.A., Sandeman, H., Peterson, T.D., LeCheminant, A.N., 2001. Enriched Archean lithospheric mantle beneath western Churchill Province tapped during Paleoproterozoic orogenesis. Geology 29, 827–830.

Cousens, B.L., Aspler, L.B., Chiarenzelli, J.R., 2004. Dual sources of ensimatic magmas, Hearne domain, Western Churchill Province, Nunavut, Canada: Neorchean infant arc processes? Precambrian Research 134, 169–188.

Cox, K.G., 1980. A model for flood basalt volcanism. Journal of Petrology 21, 629–650.

Davidson, A., 1970. Precambrian Geology, Kaminak Lake Map-area, District of Keewatin. Geological Survey of Canada, Paper 69-51, 27 p.

Davis, W., Hanmer, S., Sandeman, H.A., 2004. Temporal evolution of the Neoarchean Central Hearne supracrustal belt: rapid generation of juvenile crust in a suprasubduction zone setting. Precambrian Research 134, 85–112.

DePaolo, D.J., 1981a. Neodymium isotopes in the Colorado Front Range and crust–mantle evolution in the Proterozoic. Nature 291, 193–196.

DePaolo, D.J., 1981b. Trace element and isotopic effects of combined wallrock assimilation and fractional crystallization. Earth and Planetary Science Letters 53, 189–202.

Downes, H., 2001. Formation and modification of the shallow subcontinental lithospheric mantle; a review of geochemical evidence from ultramafic xenolith suites and tectonically emplaced ultramafic massifs of western and central Europe. Journal of Petrology 42, 233–250.

Dupuy, C., Dostal, J., 1984. Trace element geochemistry of some continental tholeiites. Earth and Planetary Science Letters 67, 61–69.

Dupuy, C., Michard, A., Dostal, J., Dautel, D., Baragar, W.R.A., 1992. Proterozoic flood basalts from the Coppermine River area, Northwest Territories: isotope and trace element geochemistry. Canadian Journal of Earth Sciences 29, 1937–1943.

Eade, K.E., 1974. Geology of Kognak River Area, District of Keewatin, Northwest Territories. Geological Survey of Canada, Memoir 377.

Eade, K.E., 1986. Precambrian Geology of the Tulemalu Lake-Yathkyed Lake Area, District of Keewatin. Geological Survey of Canada, Paper 84-11, 31 pp.

Ellam, R.M., Cox, K.G., 1991. An interpretation of Karoo picrite basalts in terms of interaction between asthenospheric magmas and mantle lithosphere. Earth and Planetary Science Letters 105, 330–342.

Elliott, T., Plank, T., Zindler, A., White, W., Bourdon, B., 1997. Element transport from slab to volcanic front at the Mariana arc. Journal of Geophysical Research 102, 14991–15019.

Ernst, R.E., Bleeker, W., 2010. Large Igneous Provinces (LIPs), giant dyke swarms, and mantle plumes: significance for breakup events within Canada and adjacent regions from 2.5 Ga to present. Canadian Journal of Earth Science 47, 695–739.

Ernst, R.E., Buchan, K.L., 2010. Geochemical Database of Proterozoic Intraplate Mafic Magmatism in Canada. Geological Survey of Canada, Open File 6016.

Ernst, R.E., Srivastava, R.K., Bleeker, W., Hamilton, M., 2010. Precambrian Large Igneous Provinces (LIPs) and their dyke swarms: new insights from high-precision geochronology integrated with paleomagnetism and geochemistry. Precambrian Research 183 (3), vii–xi.

Foley, S., 1992. Vein-plus-wall-rock melting mechanisms in the lithosphere and the origin of potassic alkaline magmas. Lithos 28, 435–453.

Foulger, G.R., Natland, J.H., Presnall, D.C., Anderson, D.L. (Eds.), 2005. Plates, plumes, and paradigms. Geological Society of America Special Paper 388.

French, J.E., Heaman, L.M., 2010. Precise U–Pb dating of Paleoproterozoic mafic dyke swarms of the Dharwar craton, India: implications for the existence of the Neoarchean supercraton Sclavia. Precambrian Research 183, 416–441.

Gallagher, K., Hawkesworth, C.J., 1992. Dehydration melting and the generation of continental flood basalts. Nature 358, 57–59.

Garfunkel, Z., 2008. Formation of continental flood volcanism—the perspective of setting of melting. Lithos 100, 49–65.

Garland, F., Turner, S., Hawkesworth, C., 1996. Shifts in the source of Paraná basalts through time. Lithos 37, 223–243.

Gibson, S.A., Thompson, R.N., Dickin, A.P., Leonardos, O.H., 1996. Erratum to high-Ti and low-Ti mafic potassic magmas: key to plume lithosphere interaction and continental flood basalt genesis. Earth and Planetary Science Letters 141, 325–341.

Griffin, W.L., O'Reilly, S.Y., Afonso, J.C., Begg, G.C., 2008. The composition and evolution of lithospheric mantle: a re-evaluation and its tectonic implications. Journal of Petrology 50, 1185–1204.

Halls, H.C., Fahrig, W.F., 1990. Mafic Dyke Swarms, vol. 34. Geological Association of Canada Special Paper, 503 pp.

Halls, H.C., Heaman, L.M., 2000. The paleomagnetic significance of new U–Pb age data from the Molson dyke swarm, Cauchon Lake area, Manitoba. Canadian Journal of Earth Sciences 37, 957–966.
Halls, H.C., Stott, G.M., Davis, D.W., 2005. Paleomagnetism, geochronology and geochemistry of several Proterozoic mafic dike swarms in northwestern Ontario. Ontario Geological Survey. Open File Report 6171, 59.
Halls, H.C., Davis, D.W., Stott, G.M., Ernst, R.E., Hamilton, M.A., 2008. The Paleoproterozoic Marathon Large Igneous Province: new evidence for a 2.1 Ga long-lived mantle plume event along the southern margin of the North American Superior Province. Precambrian Research 162, 327–353.
Hamilton, M.A., 2009. Datation isotopique (U–Pb) d'un diabase de l'essaim de dykes Mistassini, Quebec—U–Pb isotopic dating of a diabase dyke of the Mistassini swarm. QERDEM—Dossier d'exploration miniere, Documents internes. GM 65972, Quebec, 13 p.
Hanmer, S., Sandeman, H.A., Davis, W.J., Aspler, L.B., Rainbird, R.H., Peterson, T.D., Ryan, J.J., Roest, W.R., Relf, C., Irwin, D., 2004. Geology and Neoarchean tectonic setting of the Central Hearne supracrustal belt, Western Churchill Province, Nunavut, Canada. Precambrian Research 134, 63–83.
Hart, W.K., WoldeGabriel, G., Walter, R.C., Mertzman, S.A., 1989. Basaltic volcanism in Ethiopia: constraints on continental rifting and mantle interactions. Journal of Geophysical Research 94, 7731–7748.
Hawkesworth, C.J., Kempton, P.D., Rogers, N.W., Ellam, R.M., van Calsteren, P.W., 1990. Continental mantle lithosphere, and shallow level enrichment processes in the Earth's mantle. Earth and Planetary Science Letters 96, 256–268.
Heaman, L.M., 1994. 2.45 Ga global mafic magmatism: Earth's oldest superplume? In: Lanphere, M.A., Dalrymple, G.B., Turrin, B.D. (Eds.), Abstracts of the 8th International Conference on Geochronology, Cosmochronology and Isotope Geology. United States Geological Survey, Circular 1107, p. 132.
Heaman, L.M., 1997. Global mafic magmatism at 2.45 Ga: remnants of an ancient large igneous province? Geology 25, 299–302.
Heaman, L.M., LeCheminant, A.N., 1993. Paragenesis and U–Pb systematics of baddeleyite (ZrO_2). Chemical Geology 110, 95–126.
Heaman, L.M., Corkery, M.T., 1996. Proterozoic tectonic evolution of the eastern Trans Hudson margin: preliminary U–Pb results from the Split Lake Block, Manitoba. Geological Association of Canada Program with Abstracts 21, A43.
Heaman, L.M., Creaser, R.A., Cookenboo, H.O., Chacko, T., 2006. Multi-stage modification of the northern Slave mantle lithosphere: evidence from zircon- and diamond-bearing eclogite xenoliths entrained in Jericho kimberlite, Canada. Journal of Petrology 47, 821–856.
Heaman, L.M., Peck, D., Toope, K., 2009. Timing and geochemistry of 1.88 Ga Molson Igneous Events, Manitoba: insights into the formation of a craton-scale magmatic and metallogenic province. Precambrian Research 172, 143–162.
Henderson, J.R., 1981. Strain Ratios Measured Across a Hudsonian Deformation Front at Kaminak Lake, District of Keewatin. Geological Survey of Canada, Paper 1981-1A, p. 384.
Hergt, J.M., Peate, D.W., Hawkesworth, C.J., 1991. The petrogenesis of Mesozoic Gondwana low-Ti flood basalts. Earth and Planetary Science Letters 105, 134–148.
Irvine, T.N., Barager, W.R.A., 1971. A guide to the chemical classification of the common volcanic rocks. Canadian Journal of Earth Sciences 8, 523–548.
Jacques, A.L., Lewis, J.D., Smith, C.B., Gregory, G.P., Ferguson, J., Chappell, B.W., McCulloch, M.T., 1986. The diamond-bearing ultrapotassic (lamproitic) rocks of the west Kimberly region, western Australia. In: Kornprobst, J. (Ed.), Kimberlites 1. kimberlites and related rocks. Elsevier, Amsterdam, pp. 225–254.
Jaffey, A.H., Flynn, K.F., Glendenin, L.E., Bentley, W.C., Essling, A.M., 1971. Precision measurements of half-lives and specific activities of ^{235}U and ^{238}U. Physics Review C4, 1889–1906.
Jensen, L.S., 1976. A New Cation Plot for Classifying Subalkalic Volcanic Rocks. Ontario Division of Mines, Miscellaneous Paper 66, 22 pp.
Kieffer, B., Arndt, N., Lapierre, H., Bastien, F., Bosch, D., Pecher, A., Yirgu, G., Ayalew, D., Weia, D., Jerram, D.A., Keller, F., Meugniot, C., 2004. Flood and shield basalts from Ethiopia: magmas from the African superswell. Journal of Petrology 45, 793–834.
Klausen, M.B., Söderlund, U., Olsson, J.R., Ernst, R.E., Armoogam, M., Mkhize, S.W., Petzer, G., 2010. Petrological discrimination among Precambrian dyke swarms: Eastern Kaapvaal craton (South Africa). Precambrian Research 183, 501–522.
Kretz, R., 1983. Symbols for rock-forming minerals. American Mineralogist 68, 277–279.
Kullerud, K., Skjerlie, K.P., Corfu, F., de la Rosa, J.D., 2006. The 2.40 Ga Ringvassøy mafic dykes, West Troms Basement Complex, Norway: the concluding act of early Palaeoproterozoic continental breakup. Precambrian Research 150, 183–200.
Langmuir, C., Vocke, R., Hanson, G., Hart, S., 1978. A general mixing equation with applications to Icelandic basalts. Earth and Planetary Science Letters 37, 380–392.
LeBas, M.J., LeMaitre, R.W., Streckeisen, A., Zanettin, B., 1986. A chemical classification of volcanic rocks based on the total alkali silica diagram. Journal of Petrology 27, 745–750.
LeCheminant, A.N., Heaman, L.M., 1989. Mackenzie igneous events, Canada: middle Proterozoic hotspot magmatism associated with ocean opening. Earth and Planetary Science Letters 96, 38–48.
LeCheminant, A.N., Heaman, L.M., van Breemen, O., Ernst, R.E., Baragar, W.R.A., Buchan, K.L., 1996. Mafic magmatism, mantle roots and kimberlites in the Slave craton. In: LeCheminant, A.N., Richardson, D.G., DiLabio, R.N.W., Richardson, K.A. (Eds.), Searching for Diamonds in Canada. Geological Survey of Canada, pp. 161–169, Open File 3228.
LeCheminant, A.N., van Breemen, O., 1994. U–Pb Ages of Proterozoic Dyke Swarms, Lac de Gras Area, N.W.T.: Evidence for Progressive Break-up of An Archean Supercontinent. Geological Association of Canada/Mineralogical Association of Canada, Program with Abstracts, v. 19, p. A62.
Lehnert, K., Su, Y., Langmuir, C., Sarbas, B., Nohl, U., 2000. A global geochemical database structure for rocks. Geochemistry Geophysics Geosystems 1, Paper number 1999GC000026.
Ludwig, K.R., 2003. ISOPLOT, A Plotting and Regression Program for Radiogenic Isotope Data, Version Ex/3.00. Berkeley Geochronology Center Special Publication.
Lugmair, G.W., Marti, K., 1978. Lunar initial $^{143}Nd/^{144}Nd$: differential evolution of the lunar crust and mantle. Earth and Planetary Science Letters 39, 3349–3357.
Mackie, R.A., Scoates, J.S., Weis, D., 2009. Age and Nd–Hf isotopic constraints on the origin of marginal rocks from the Muskox layered intrusion (Nunavut, Canada) and implications for the evolution of the 1.27 Ga Mackenzie large igneous province. Precambrian Research 172, 46–66.
Mahoney, J.J., Coffin, M.F. (Eds.), 1997. Large Igneous Provinces Continental, Oceanic, and Planetary Flood Volcanism. Geophysical Monograph 100, American Geophysical Union.
Maurice, C., David, J., O'Neil, J., Francis, D., 2009. Age and tectonic implications of Paleoproterozoic mafic dyke swarms for the origin of 2.2 Ga enriched lithosphere beneath the Ungava Peninsula, Canada. Precambrian Research 174, 163–180.
McKenzie, D., O'Nions, R.K., 1995. The source regions of ocean island basalts. Journal of Petrology 36, 133–159.
Melluso, L., Mahoney, J.J., Dallai, L., 2006. Mantle sources and crustal input as recorded in high-Mg Deccan Traps basalt of Gujarat (India). Lithos 89, 259–274.
Mertanen, S., Halls, H.C., Vuollo, J.I., Pesonen, L.J., Stepanov, V.S., 1999. Paleomagnetism of 2.44 Ga mafic dykes in Russian Karelia, eastern Fennoscandian Shield—implications for continental reconstructions. Precambrian Research 98, 197–221.
Middelburg, J.J., Van der Weijden, C.H., Woittiez, J.R.W., 1988. Chemical processes affecting the mobility of major, minor and trace elements during weathering of granitic rocks. Chemical Geology 68, 253–273.
Mitchell, R.H., Bergman, S.C., 1991. Petrology of Lamproites. Plenum Publishing, New York.
Miyashiro, A., 1974. Volcanic rock series in island arcs and active continental margins. American Journal of Science 274, 321–355.
Nelson, D.O., Morrison, D.A., Phinney, W.C., 1990. Open-system evolution versus source control in basaltic magmas: Matachewan dyke swarm, Superior Province, Canada. Canadian Journal of Earth Sciences 27, 767–783.
Paces, J.B., Bell, K., 1989. Non-depleted sub-continental mantle beneath the Superior Province of the Canadian Shield: Nd–Sr and trace element evidence from midcontinental rift basalts. Geochimica et Cosmochimica Acta 53, 2023–2035.
Park, A.F., Ralser, S., 1992. Precambrian Geology of the southwestern part of the Tavani map area, District of Keewatin, Northwest Territories. Geological Survey of Canada Bulletin 416, 81p.
Patterson, J.G., 1991. The Spi Group: a post-Archean, pre-2.1 Ga rift succession, Trans-Hudson hinterland. Canadian Journal of Earth Sciences 28, 1863–1872.
Pearce, J.A., 1982. Trace element characteristics of lavas from destructive plate boundaries. In: Thorpe, R.S. (Ed.), Andesites. John Wiley and Sons, pp. 525–548.
Pearce, J.A., 1996. A user's guide to basalt discrimination diagrams. Trace Element Geochemistry of Volcanic Rocks; Applications for Massive Sulphide Exploration. Short Course Notes, vol. 12. Geological Association of Canada, pp. 79–113.
Pearce, J.A., 2008. Geochemical fingerprinting of oceanic basalts with applications to ophiolite classification and the search for Archean oceanic crust. Lithos 100, 14–48.
Pearce, J.A., Cann, J.R., 1973. Tectonic setting of basic volcanic rocks determined using trace element analyses. Earth and Planetary Science Letters 19, 290–300.
Pearce, J.A., Norry, M.J., 1979. Petrogenetic implications of Ti, Zr, Y and Nb variations in volcanic rocks. Contributions to Mineralogy and Petrology 69, 33–47.
Peate, D.W., Hawkesworth, C.J., 1996. Lithospheric to asthenospheric transition in low-Ti flood basalts from southern Paraná, Brazil. Chemical Geology 127, 1–24.
Peterson, T.D., 1997. New mapping of Archean supracrustal rocks in the Kaminak Belt, District of Keewatin, Northwest Territories. Geological Survey of Canada, Current Research 1997-C, 111–121.
Peterson, T.D., Pehrsson, S., Skulski, T., Sandeman, H., 2010. Compilation of Sm–Nd Isotope Analyses of Igneous Suites, Western Churchill Province. Geological Survey of Canada, Open File 6439, 18p.
Phinney, W.C., Halls, H.C., 2001. Petrogenesis of the Early Proterozoic Matachewan dyke swarm, Canada, and implications for magma emplacement and subsequent deformation. Canadian Journal of Earth Sciences 38, 1541–1563.
Presnall, D.C., Gudfinnsson, G.H., 2011. Oceanic volcanism from the low-velocity zone without mantle plumes. Journal of Petrology 52, 1533–1546.
Rainbird, R.H., Davis, W.J., Aspler, L.B., Chiarenzelli, J.R., Ryan, J.J., 2002. SHRIMP U–Pb detrital zircon geochronology of enigmatic Neoarchean–Paleoproterozoic sedimentary rocks of the central western Churchill Province, Nunavut. In: Radiogenic Age and Isotopic Studies; Report 15, Geological Survey of Canada, Current Research 2002-F5.
Ray, R., Sheth, H.C., Mallik, J., 2007. Structure and emplacement of the Nandurbar-Dhule mafic dyke swarm, Deccan Traps, and the tectonomagmatic evolution of flood basalts. Bulletin of Volcanology 69, 537-531.
Ridler, R.H., 1974. Shallow Marine Plateau Basalts of the Aphebian Hurwitz Group at Last Lake, District of Keewatin. Geological Survey of Canada, Paper 74-1, Part B, pp. 195–199.
Ridler, R.H., Shilts, W.W., 1974. Mineral potential of the Rankin Inlet, Ennadai Belt. Canadian Mining Journal 95, 32–42.
Rock, N.M.S., 1991. Lamprophyres. Van Nostrand Reinhold, New York, 285 p.

Rollinson, H.R., 1993. Using Geochemical Data: Evaluation, Presentation and Interpretation. Longman Scientific and Technical Press, Harlow, UK.
Rudnick, R.L., Gao, S., 2006. Composition of the continental crust. Treatise On Geochemistry, vol. 3, pp. 1–64, ISBN: 0-08-044338-9.
Sandeman, H.A., 2005. Petrochemistry of Proterozoic continental tholeiites of the Western Churchill Province and implications for the evolution of the subcontinental lithosphere. In: Geological Association of Canada–Mineralogical Association of Canada Joint Annual Meeting, Halifax.
Sandeman, H.A., 2007. Secular and spatial variations in the composition of Paleoproterozoic (2200–1950 Ma), Manikewan basaltic suites of the western Churchill Province: implications for regional lithostratigraphic reconstructions and evolution of mantle sources. In: Geological Association of Canada–Mineralogical Association of Canada Joint Annual Meeting, Yellowknife.
Sandeman, H.A., Ryan, J.J., 2008. The Spi Lake Formation of the central Hearne domain, western Churchill Province, Canada: an axial intracratonic continental tholeiite trough above the cogenetic Kaminak dyke swarm. Canadian Journal of Earth Sciences 45, 745–767.
Sandeman, H.A., Cousens, B.L., Hemmingway, C.J., 2003. Continental tholeiitic mafic rocks of the Paleoproterozoic Hurwitz Group Central Hearne Domain, Nunavut: insight into the evolution of the Hearne sub-continental lithosphere. Canadian Journal of Earth Sciences 40, 1219–1237.
Sandeman, H., Cousens, B., Peterson, T., Hemmingway, C., Davis, W., Hanmer, S., Tella, S., Ryan, J., Aspler, L., 2000. Petrochemistry and Nd isotopic evolution of Proterozoic mafic rocks of the Western Churchill Province, Nunavut and their implications for the evolution of the lithospheric mantle. In: Geological Association of Canada–Mineralogical Association of Canada Joint Annual Meeting, Calgary.
Sandeman, H.A., Hanmer, S., Davis, W.J., Ryan, J.J., Peterson, T.D., 2004a. Whole-rock and Nd isotopic geochemistry of Neoarchean granitoids and their bearing on the evolution of the Central Hearne supracrustal belt, Western Churchill Province, Nunavut, Canada. Precambrian Research 134, 143–167.
Sandeman, H.A., Hanmer, S., Davis, W.J., Ryan, J.J., Peterson, T.D., 2004b. Neoarchean volcanic rocks, Central Hearne supracrustal belt, Western Churchill Province, Canada: geochemical and isotopic evidence supporting intra-oceanic, suprasubduction zone extension. Precambrian Research 134, 113–141.
Schwab, D.L., Thorkelson, D.J., Mortensen, J.K., Creaser, R.A., Abbott, J.G., 2004. The Bear River dykes (1265–1269 Ma); westward continuation of the Mackenzie dyke swarm into Yukon, Canada. Precambrian Research 133, 175–186.
Sheraton, J.W., Oliver, R.L., Stuwe, K., 1989. Geochemistry of Proterozoic amphibolite dykes of Commonwealth Bay, Antarctica, and possible correlations with mafic dyke swarms elsewhere in Gondwanaland. Precambrian Research 44, 353–361.
Stacey, J.S., Kramers, J.D., 1975. Approximation of terrestrial lead isotope evolution by a two-stage model. Earth and Planetary Science Letters 26, 207–221.
Stauffer, M.R., 1984. Manikewan: an Early Proterozoic ocean in central Canada, its igneous history and orogenic closure. Precambrian Research 25, 257–281.
Stein, M., Goldstein, S.L., 1996. From plume head to continental lithosphere in the Arabian–Nubian shield. Nature 382, 773–778.
Stern, R.A., Berman, R.G., 2000. Monazite U–Pb and Th–Pb geochronology by ion microprobe, with application to in situ dating of an Archean metasedimentary rock. Chemical Geology 172, 113–130.
Stern, R.A., Hanson, G.N., 1991. Archean high-Mg granodiorite: a derivative of light rare earth element-enriched monzodiorite of mantle origin. Journal of Petrology 31, 201–238.
Sun, S.-S., McDonough, W.F., 1989. Chemical and isotopic systematics of oceanic basalts: implications for mantle composition and processes. In: Saunders, A.D., Norry, M.J. (Eds.), Magmatism in the Ocean Basins, vol. 42. Geological Society Special Publication, pp. 313–345.
Taylor, S.R., McLennan, S.M., 1985. The Continental Crust: Its Composition and Evolution. Blackwell Scientific Publications, Oxford.
Tella, S., Annesley, I.R., Borradaile, G.J., Henderson, J.R., 1986. Precambrian Geology of Tavani, Marble Island and Chesterfield Inlet Map Areas, District of Keewatin—A Progress Report. Geological Survey of Canada, Paper 86-13, 20 p.
Thériault, R.J., 1990. Methods for Rb–Sr and Sm–Nd Isotopic Analyses at the Geochronology Laboratory. Geological Survey of Canada, Paper 89-2, pp. 3–6.
Thompson, R.N., Morrison, G.L., Dickin, A.P., Hendry, G.L., 1983. Continental flood basalts. . .arachnids rule OK? In: Hawkesworth, C.J., Norry, M.J. (Eds.), Continental Basalts and Mantle Xenoliths. Shiva Geology Series, pp. 158–185.
Turner, S., Hawkesworth, C., 1995. The nature of the sub-continental mantle: constraints from the major element composition of continental flood basalts. Chemical Geology 120, 295–314.
van Breemen, O., Peterson, T.D., Sandeman, H.A., 2005. U–Pb zircon geochronology and Nd isotope geochemistry of Proterozoic granitoids in the western Churchill Province: intrusive age pattern and Archean source domains. Canadian Journal of Earth Sciences 42, 339–377.
van Keken, P.E., Hauri, E.H., Ballentine, C.J., 2002. Mantle mixing: the generation, preservation, and destruction of chemical heterogeneity. Annual Review of Earth and Planetary Sciences 30, 493–525.
Weaver, B.L., Tarney, J., 1981. The Scourie Dyke suite: petrogenesis and geochemical nature of the Proterozoic sub-continental mantle. Contributions to Mineralogy and Petrology 78, 175–188.
Williams, H., Hoffman, P.F., Lewry, J.F., Monger, J.W.H., Rivers, T., 1991. Anatomy of North America: thematic geologic portrayals of the continent. In: Hilde, T.W.G., Carlson, R.L. (Eds.), Silver Anniversary of Plate Tectonics. Tectonophysics 187, 117–134.
Wright, G.M., 1967. Geology of the southeastern barren grounds; parts of the Districts of Mackenzie and Keewatin. Geological Survey of Canada Memoir 350, 91p.